Faculty of Medicine, Constantine University3
Department of Medicine
Physiology course first year medicine

SYNAPTIC TRANSMISSION

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- **INTRODUCTION:**
- The **nerve cell** has the unique ability to communicate rapidly and precisely through:
- **Nerve conduction**
- **Synaptic transmission**
- Sherrington (1897) proposed the term **"synapse"** to designate the contact zones between neural elements.

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INTRODUCTION:

- Synapses constitute the fundamental element of communication within the nervous system.
- They ensure the transmission and modulation of neural information through complex mechanisms involving electrical, chemical, and molecular phenomena.
- The synaptic message allows for the integration and propagation of information throughout the neural network.

Definition

Synapses represent a specialized junction zone located where the axon terminal comes into contact with another neuron or another type of cell.

Synapses consist of:

The presynaptic element: generally composed of a terminal button (axon terminal).

The postsynaptic element: which can be a dendrite, the soma of another neuron, or a non-neuronal cell.

The synaptic cleft or space: the space between the presynaptic membrane and the postsynaptic membrane.

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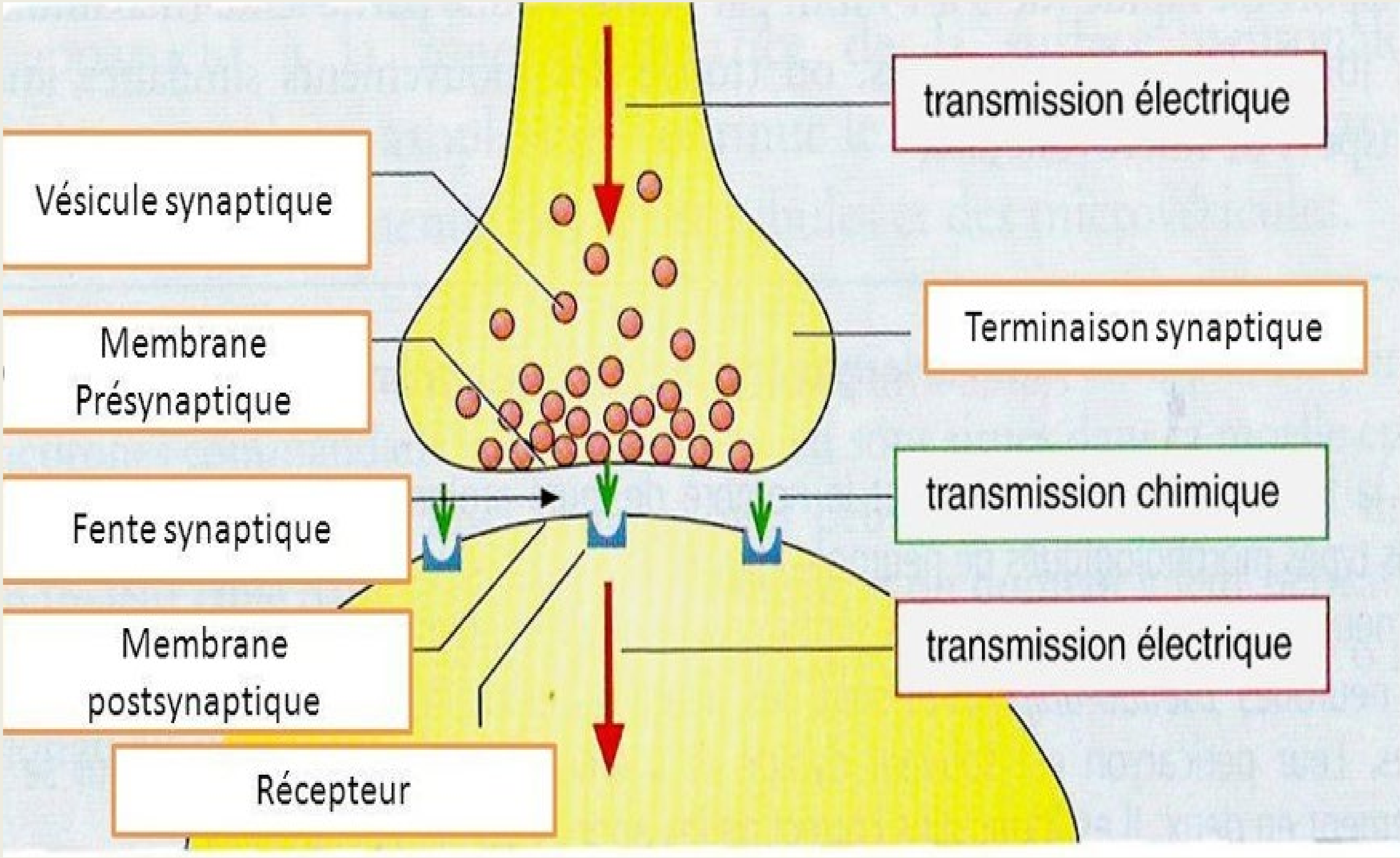
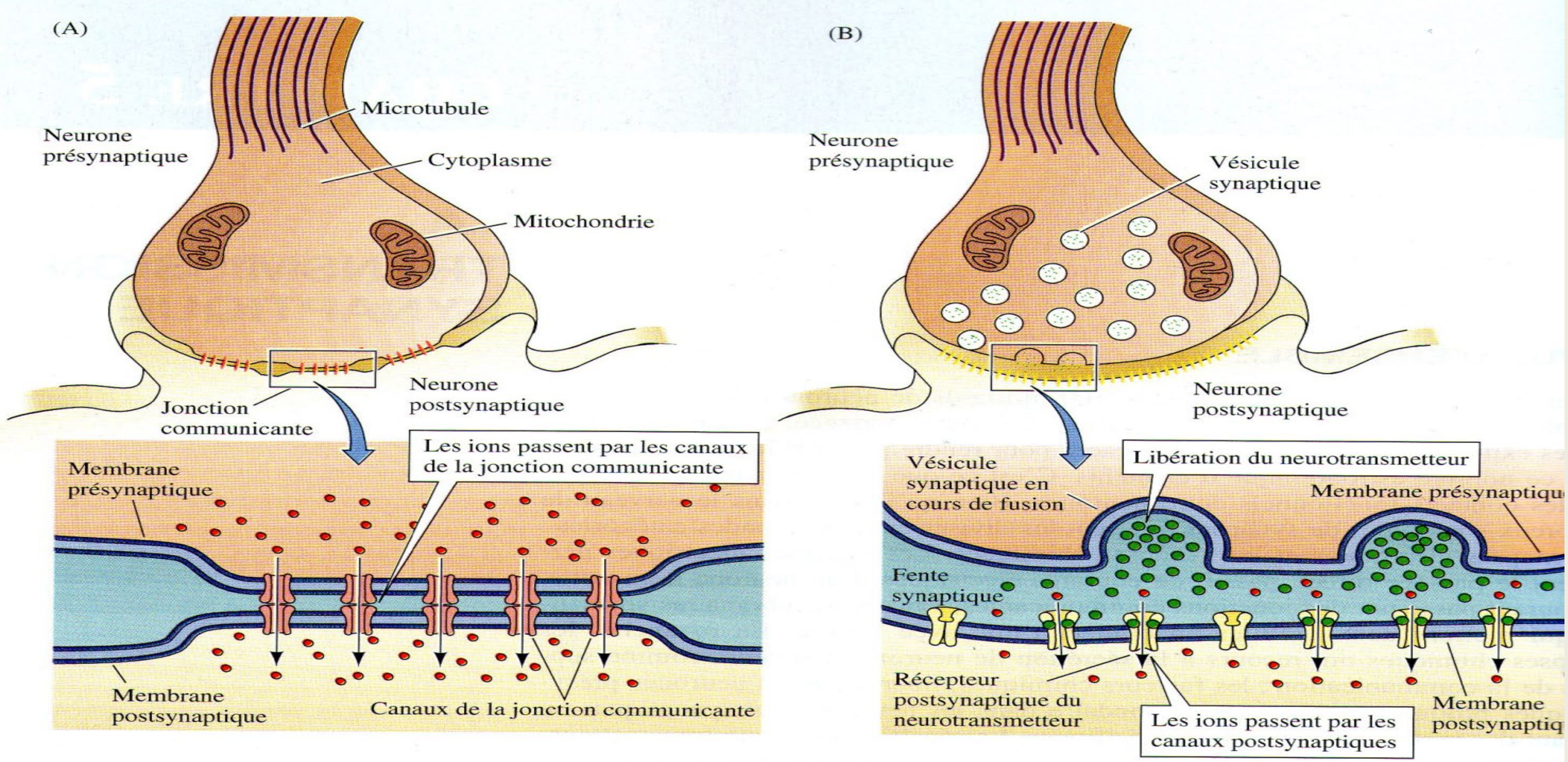
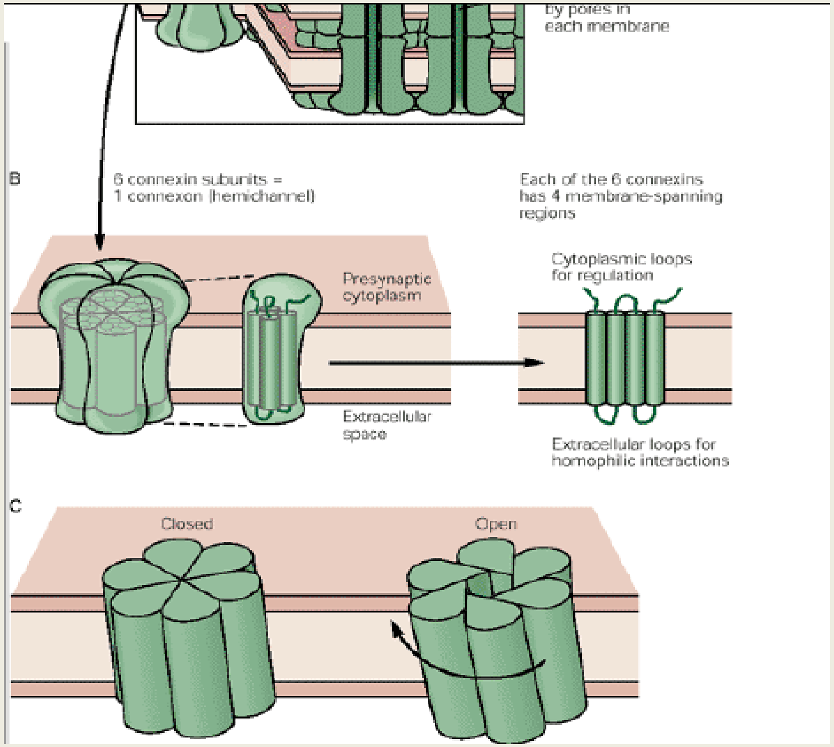
Structural and functional composition of a synapse

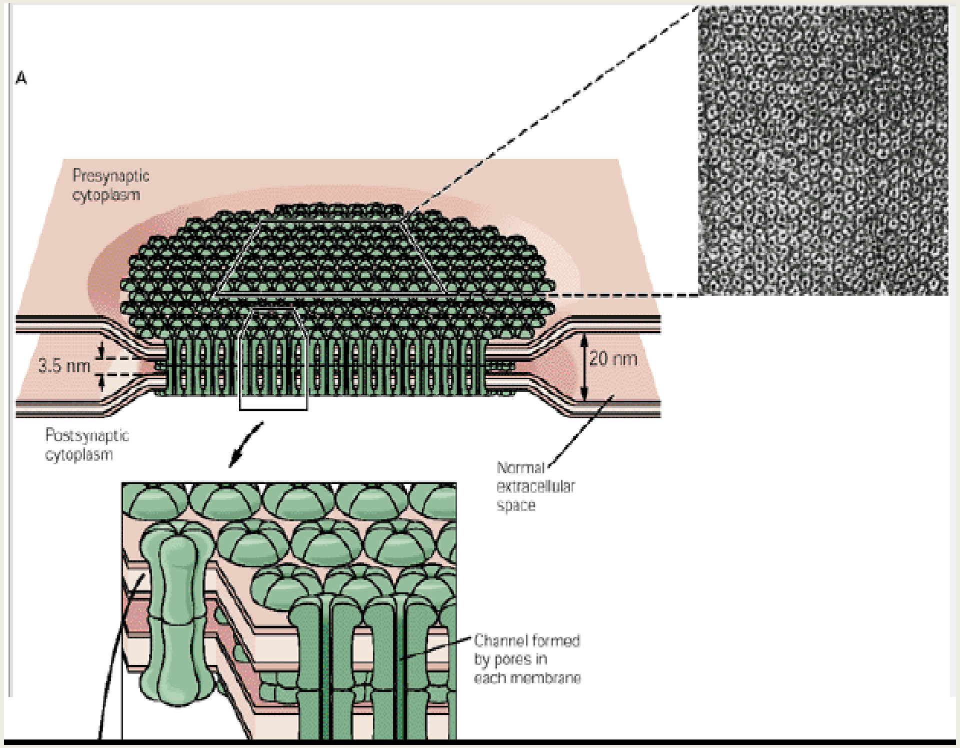


- ✓ **CLASSIFICATION**
- ✓ **Depending on the mode of operation :**
- ✓ **Electrical Synapses**
- ✓ **Chemical synapses**



SYNAPSE ELECTRIQUE

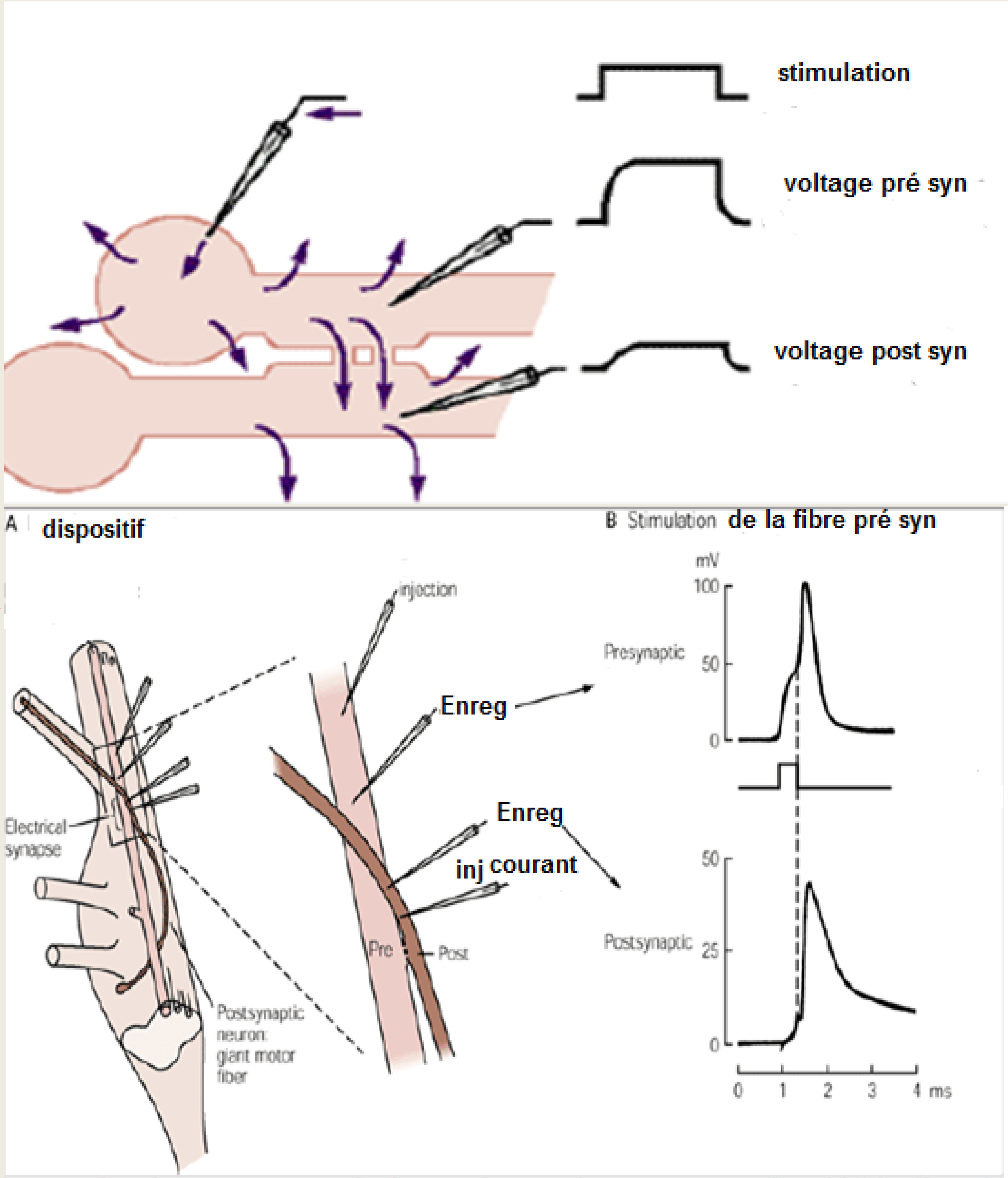




GAP JUNCTION:
Semi-channel pair: connexon
Connexon: formed of six identical subunits with terminal N and C ends
The pore at 1.5 nm diameter (ions and organic substances with low
Molecular weight)

PERMET UN COUPLAGE

METABILIC AND ELECTRICAL THROUGH "GAP JUNCTIONS" THE TRANSMISSION IS FAST, BIDIRECTIONAL AND IS DONE WITHOUT MODIFICATION



Role of the electrical synapse

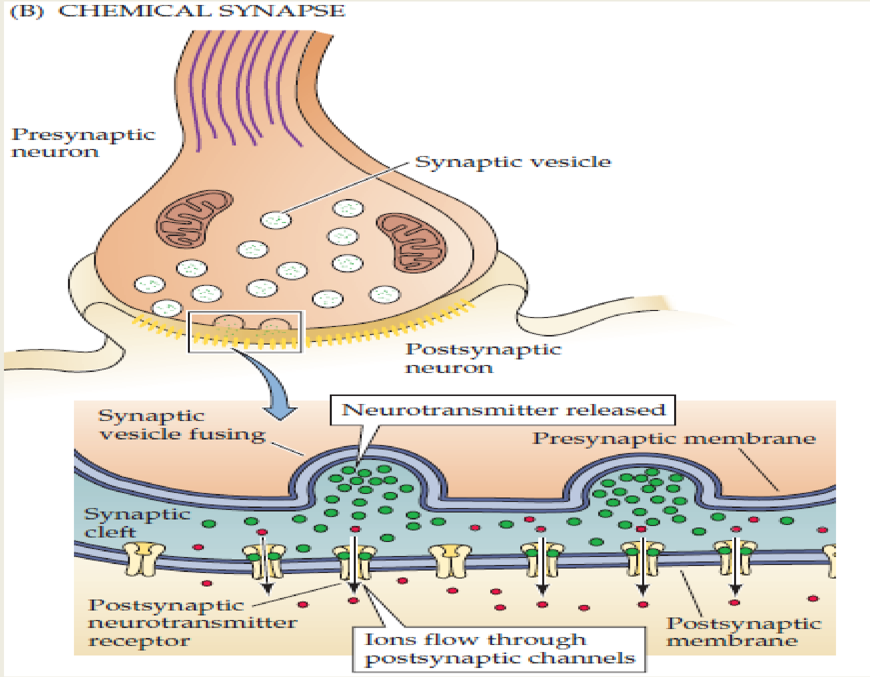
NEURAL SYNCHRONIZATION

Example: Brainstem: Regulation of respiration
Interneurons of the cerebral cortex, thalamus, cerebellum, and other regions.

CHEMICAL SYNAPSE

Need for a chemical intermediate:

(NEUROMEDIATOR / NEUROTRANSMITTER)
-Wider synaptic space
-One-way transmission, slower.



✓ CLASSIFICATION

Depending on the elements in contact:

-neuro-efector synapses: Synapses of the peripheral nervous system

between the axons of the autonomic nervous system and the glands, smooth muscle, heart and neuromuscular junction

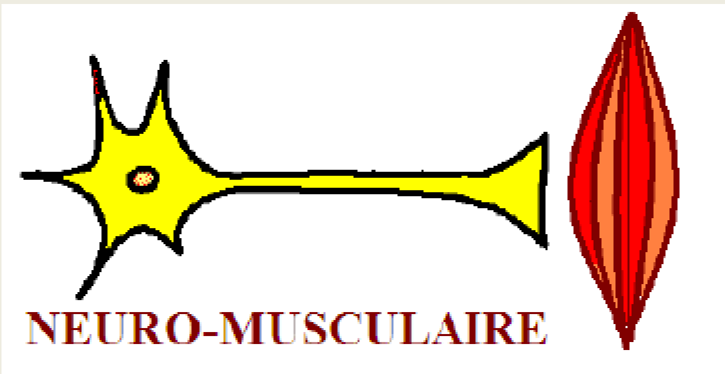
-Neuroneuronal synapses: Synapses of the central nervous system

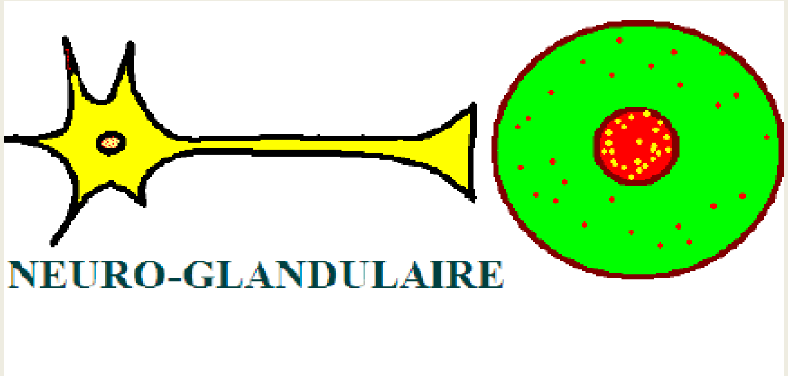
axodendritics
Axosomatic
axoaxonales
dendrodendritics

- **According to the postsynaptic effect**

Excitatory synapses
Inhibitory synapses

Nerf effector synapses:





The Chemical Synapse: The Neurotransmitter

It is a substance:

Synthesized at the level of the presynaptic neuron

Present in presynaptic terminals

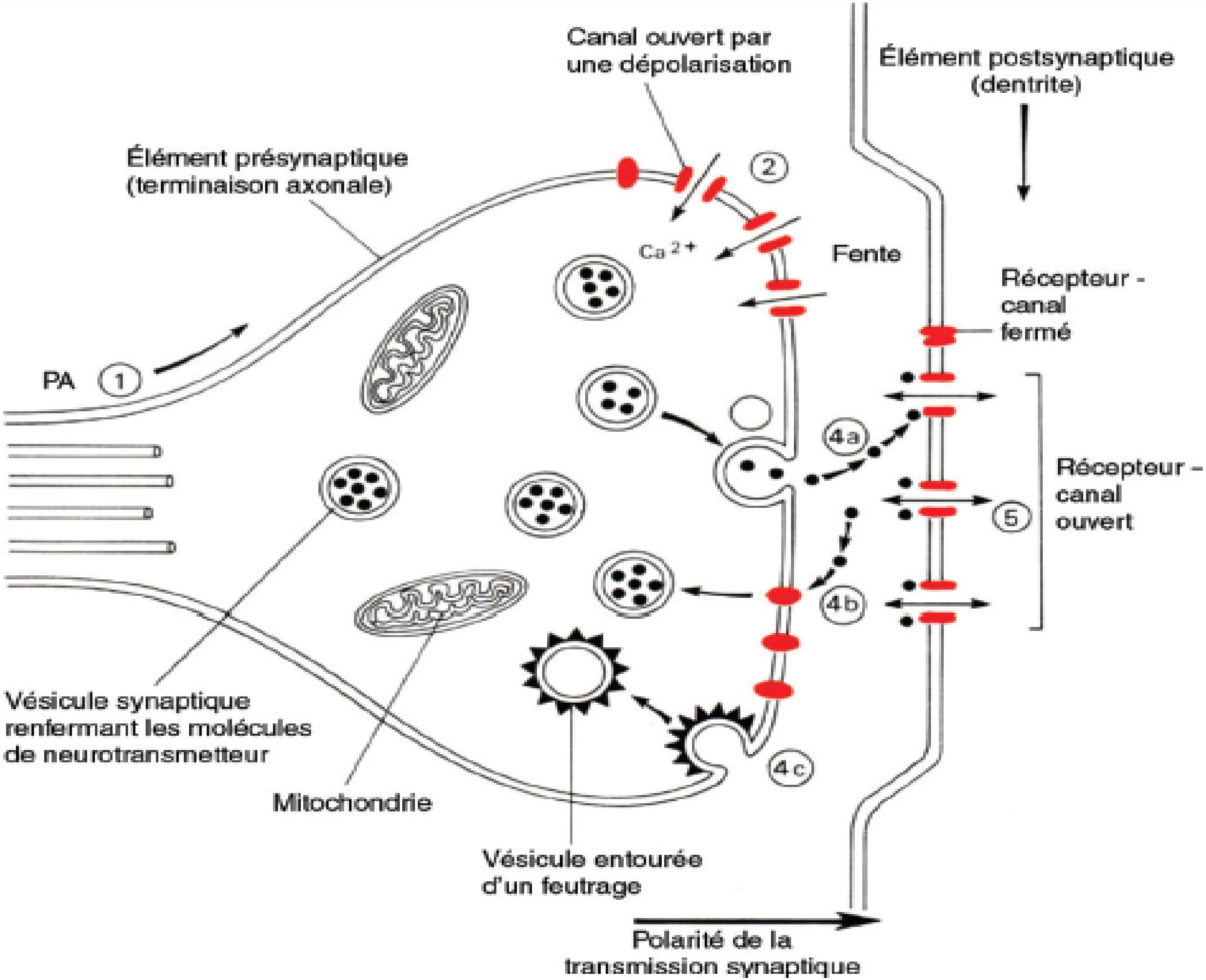
Released in sufficient quantities into the Synaptic cleft

Ca++ dependent release

Binding to specific postsynaptic receptors

Existence of inactivation mechanisms

GENERAL DIAGRAM OF HOW A CHEMICAL SYNAPSE WORKS



Presynaptic Stages

- Arrival of the Action Potential (AP) at the presynaptic terminal.
- Depolarization of the nerve terminal membranes.
- Opening of voltage-gated calcium channels.
- Influx of Ca^{2+} into the presynaptic terminals.

- The sudden increase in intracellular presynaptic Ca^{2+} concentration triggers the fusion of the presynaptic vesicle membranes with the plasma membrane of the axonal terminal.

- The presynaptic vesicles release their contents into the synaptic cleft. This process of neurotransmitter release is called exocytosis. The release of small-molecule neurotransmitters.

Post-synaptic stages

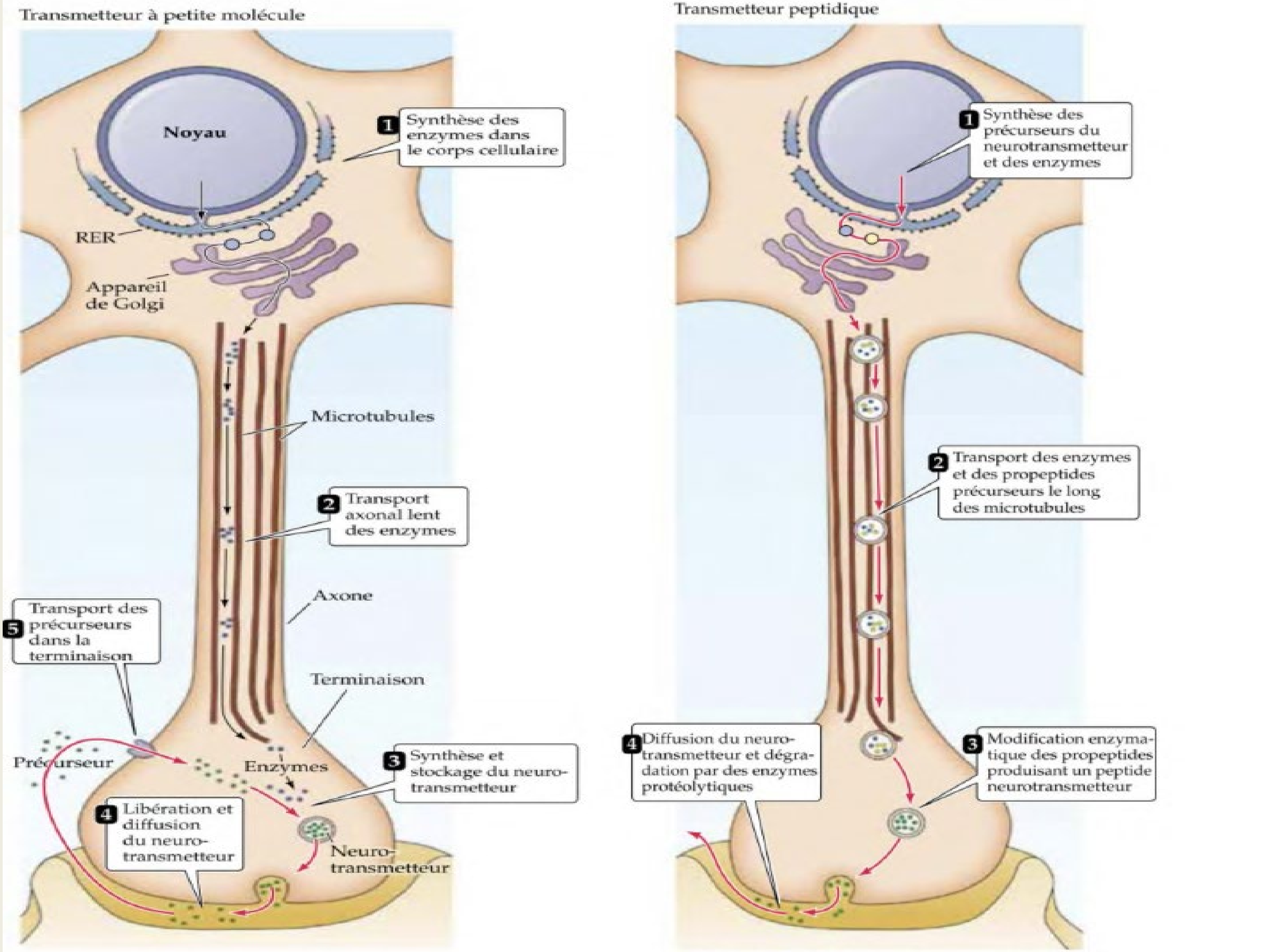
- **Neurotransmitters (NTs)** diffuse across the synaptic cleft and bind to thousands of specific receptors.
- Receptors are either **ionotropic receptors (receptor-channels) or metabotropic receptors.**
- The NT-receptor binding triggers a **conformational change** in the protein and the opening of channels in the postsynaptic membrane.

- The NT-receptor binding must then be rapidly terminated to allow for the transmission of a new chemical signal corresponding to the arrival of new action potentials.

- The NT may simply diffuse out of the synaptic cleft, be degraded within the cleft, or be reuptaken by the presynaptic cell.

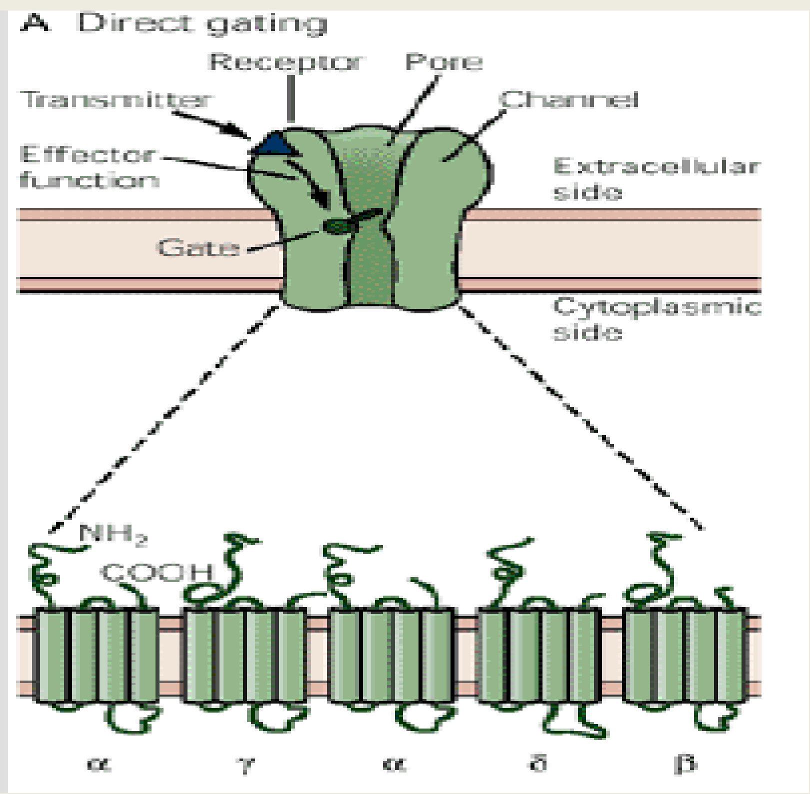
Biochemical classification of neurotransmitters

- **Small molecule neurotransmitters: amino acids amines: have a diversified physiological response**
- **Neuropeptides**

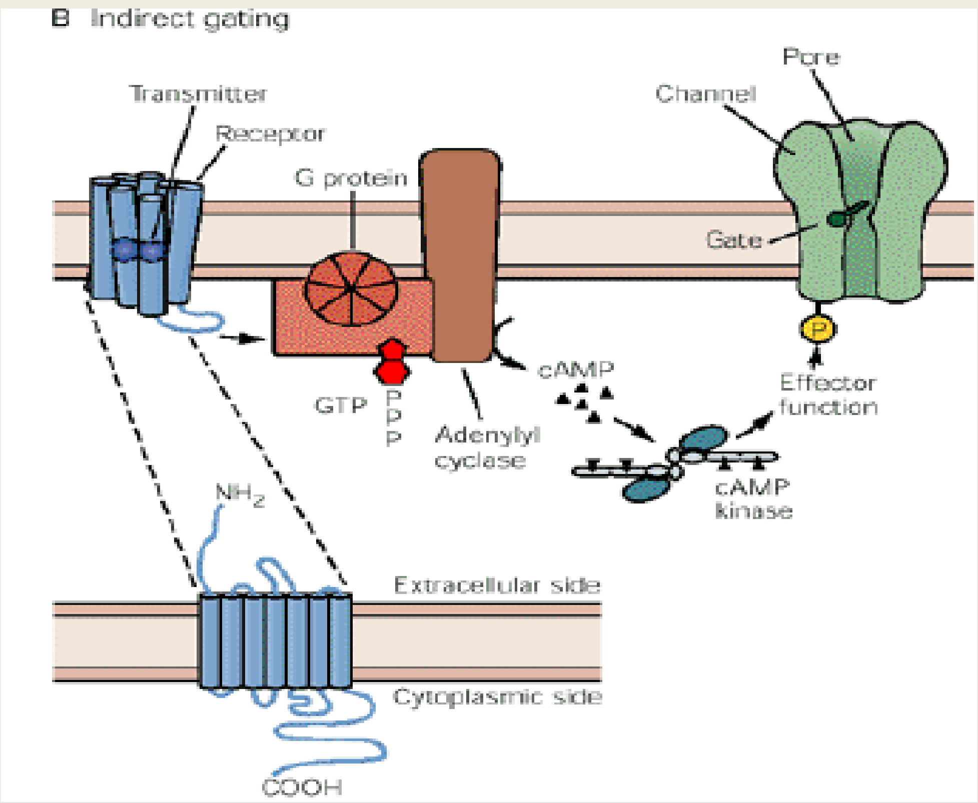


Transmetteur à petite molécule
Noyau
1 Synthèse des enzymes dans le corps cellulaire
RER
Appareil de Golgi
Microtubules
2 Transport axonal lent des enzymes
Axone
5 Transport des précurseurs dans la terminaison
Terminaison
Précurseur
Enzymes
3 Synthèse et stockage du neuro-transmetteur
4 Libération et diffusion du neuro-transmetteur
Neuro-transmetteur
Transmetteur peptidique
1 Synthèse des précurseurs du neurotransmetteur et des enzymes
2 Transport des enzymes et des propeptides précurseurs le long des microtubules
4 Diffusion du neuro-transmetteur et dégra-dation par des enzymes protéolytiques
3 Modification enzyma-tique des propeptides produisant un peptide neurotransmetteur

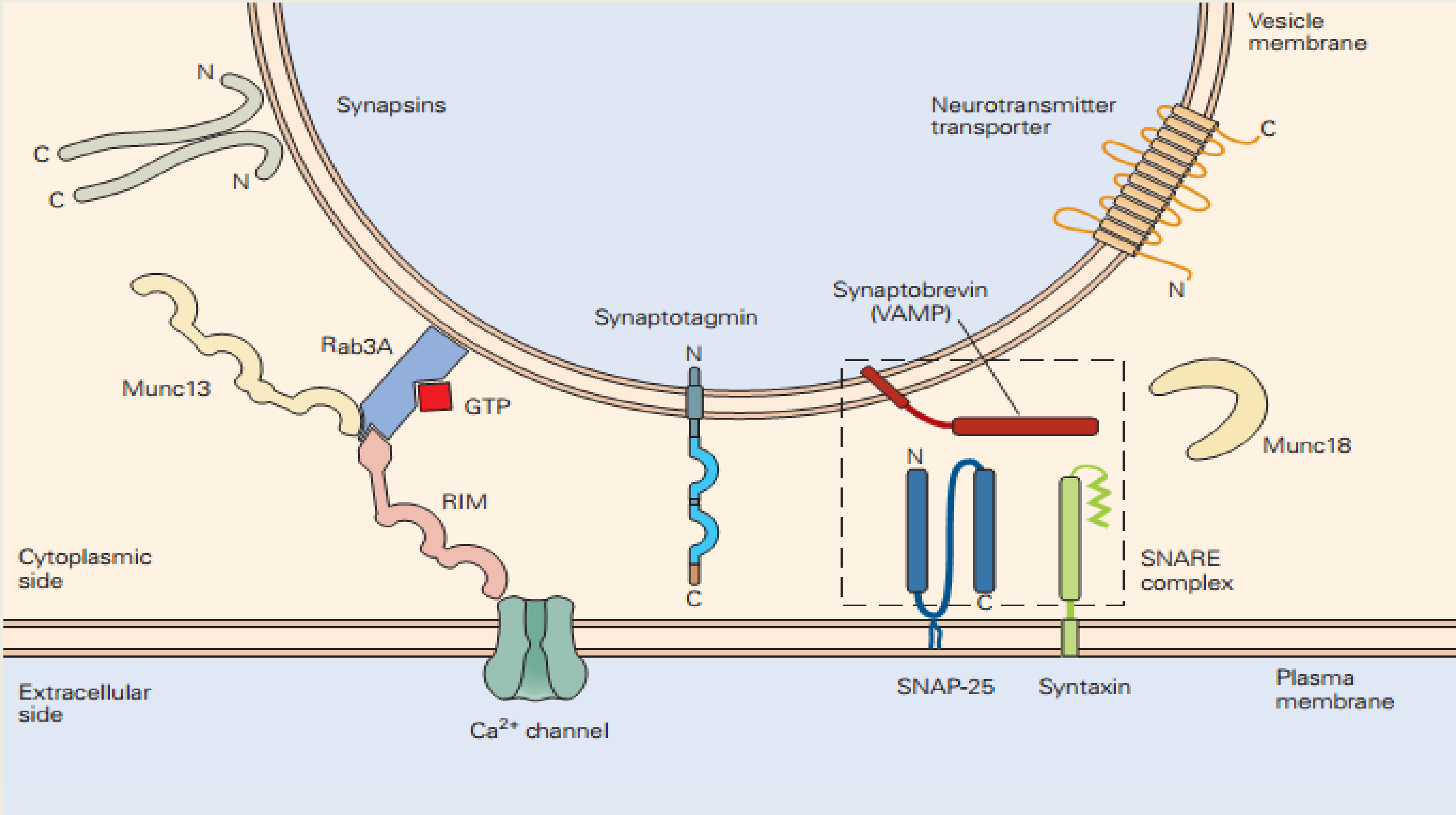
TWO TYPES OF POSTSYNAPTIC RECEPTORS



RECEPTEUR CANAL
(IONOTROPIQUE)



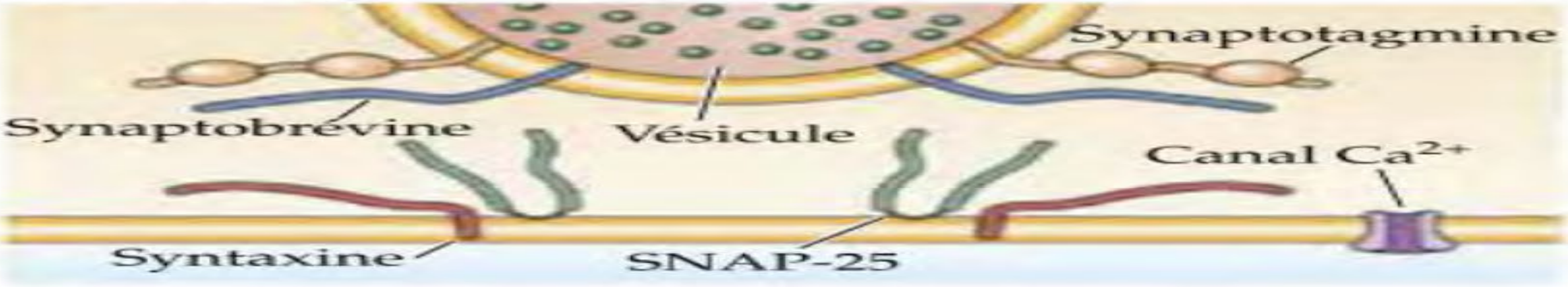
RECEPTEUR METABOTROPIQUE
(AMPLIFICATION DU SIGNAL)



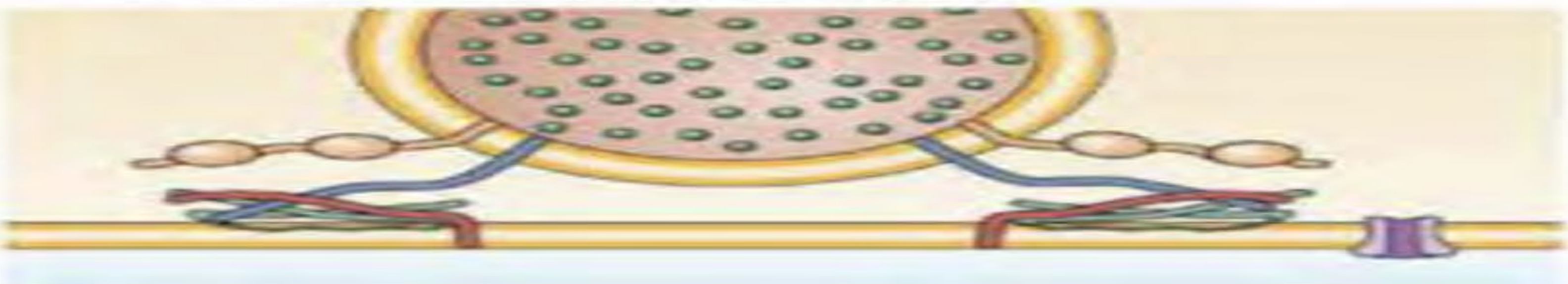
Plusieurs protéines:

- Synapsines: regulation of vesicular reserves and vesicle mobilization
- SNAREs: catalyze vesicular fusion with the plasma membrane, two types:
- - Vesicular SNARE: VAMP, synaptobrevine
- -SNARE t (target): syntaxine ,SNAP-25

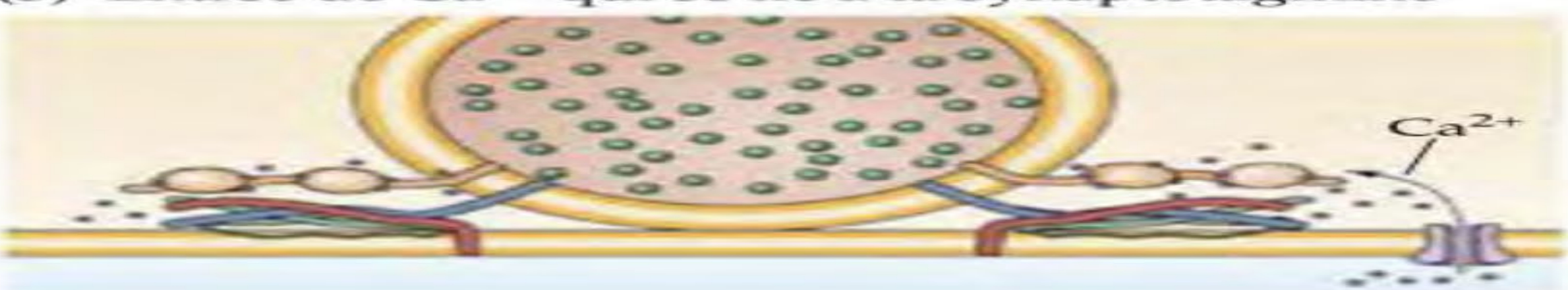
(1) Arrimage de la vésicule



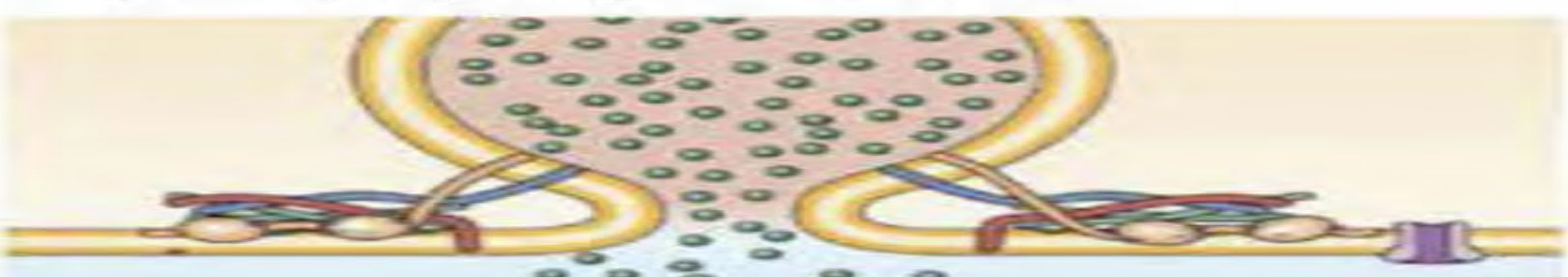
(2) Formation de complexes SNARE qui rapprochent les membranes l'une de l'autre



(3) Entrée de Ca^{2+} qui se lie à la synaptotagmine

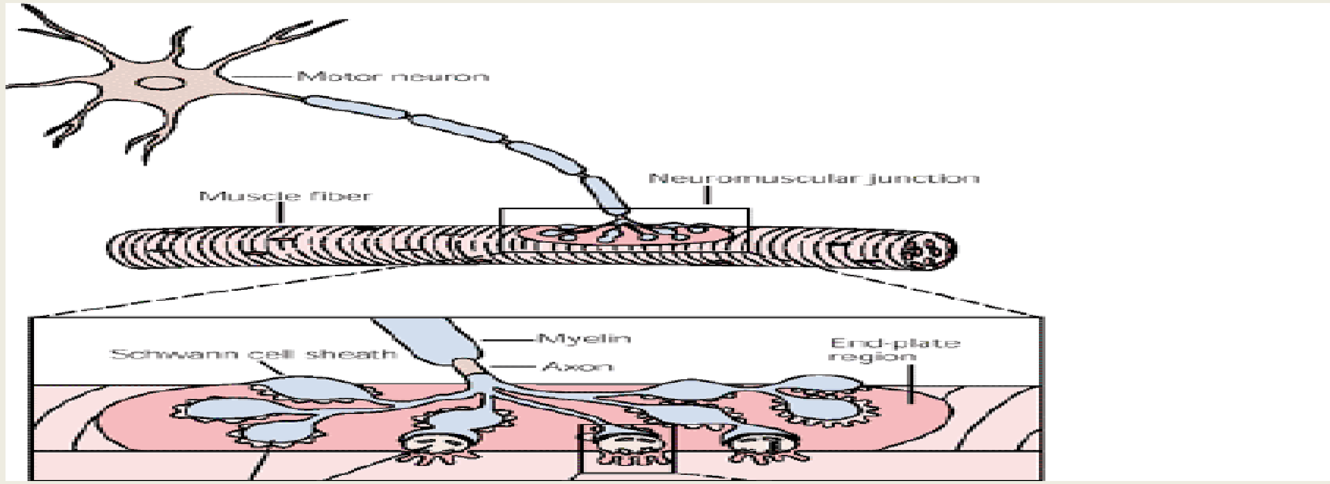


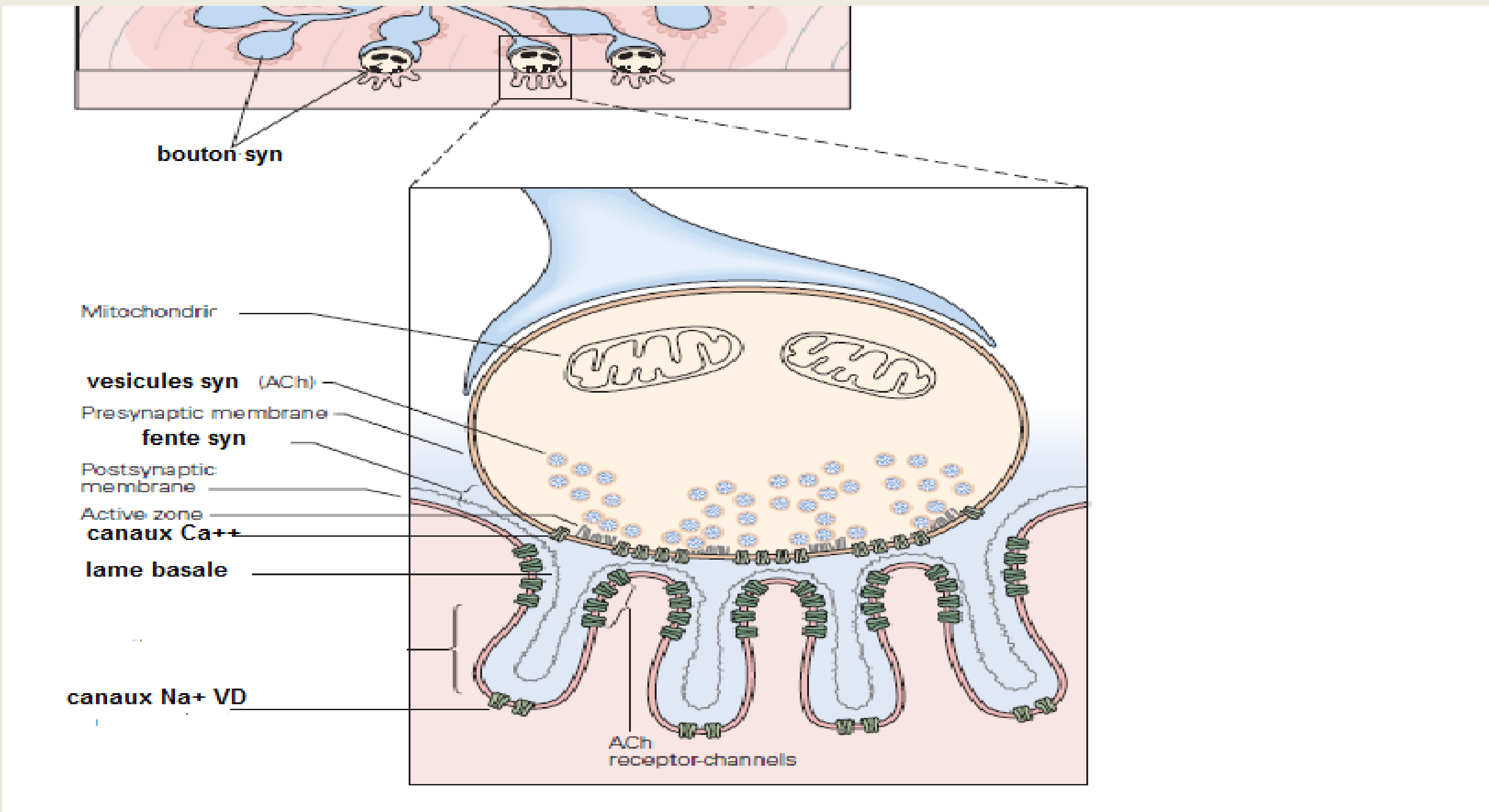
(4) Catalyse de la fusion des membranes par la synaptotagmine liée au Ca^{2+}



THE NEUROMUSCULAR JUNCTION
(motor plate)

1/Structure


Motor neuron
Neuromuscular junction
Muscle fiber
Schwann cell sheath
Myelin
Axon
End-plate
region

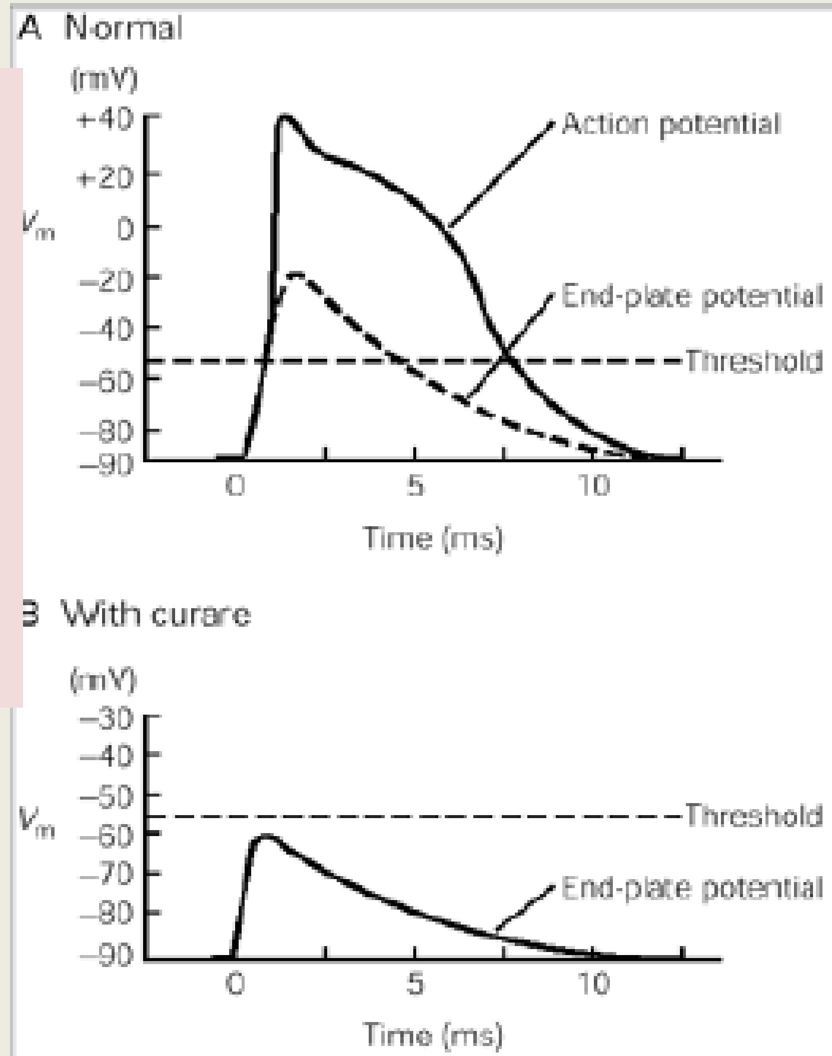

bouton syn
Mitochondrir
vesicules syn (ACh)
Presynaptic membrane
fente syn
Postsynaptic
membrane
Active zone
canaux Ca++
lame basale
canaux Na+ VD
ACh
receptor-channels

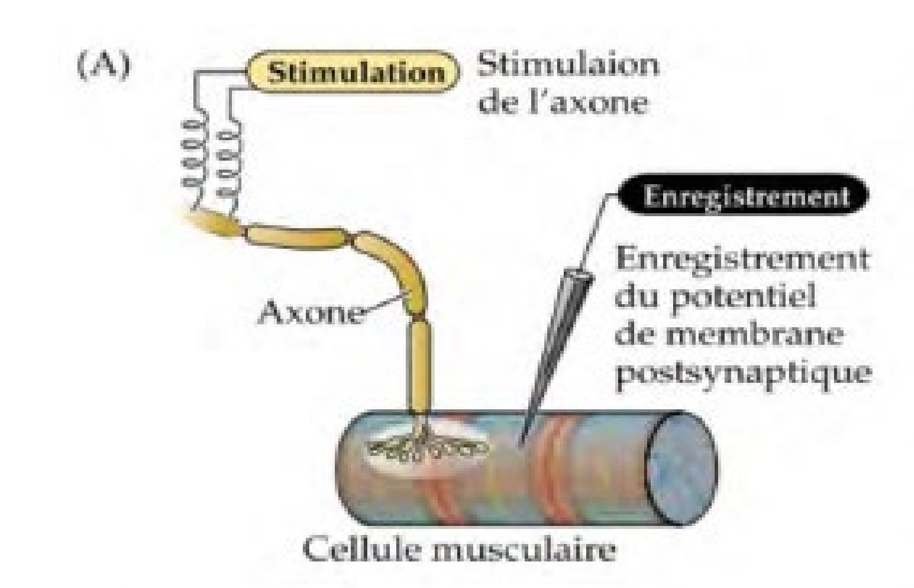
Electrophysiology

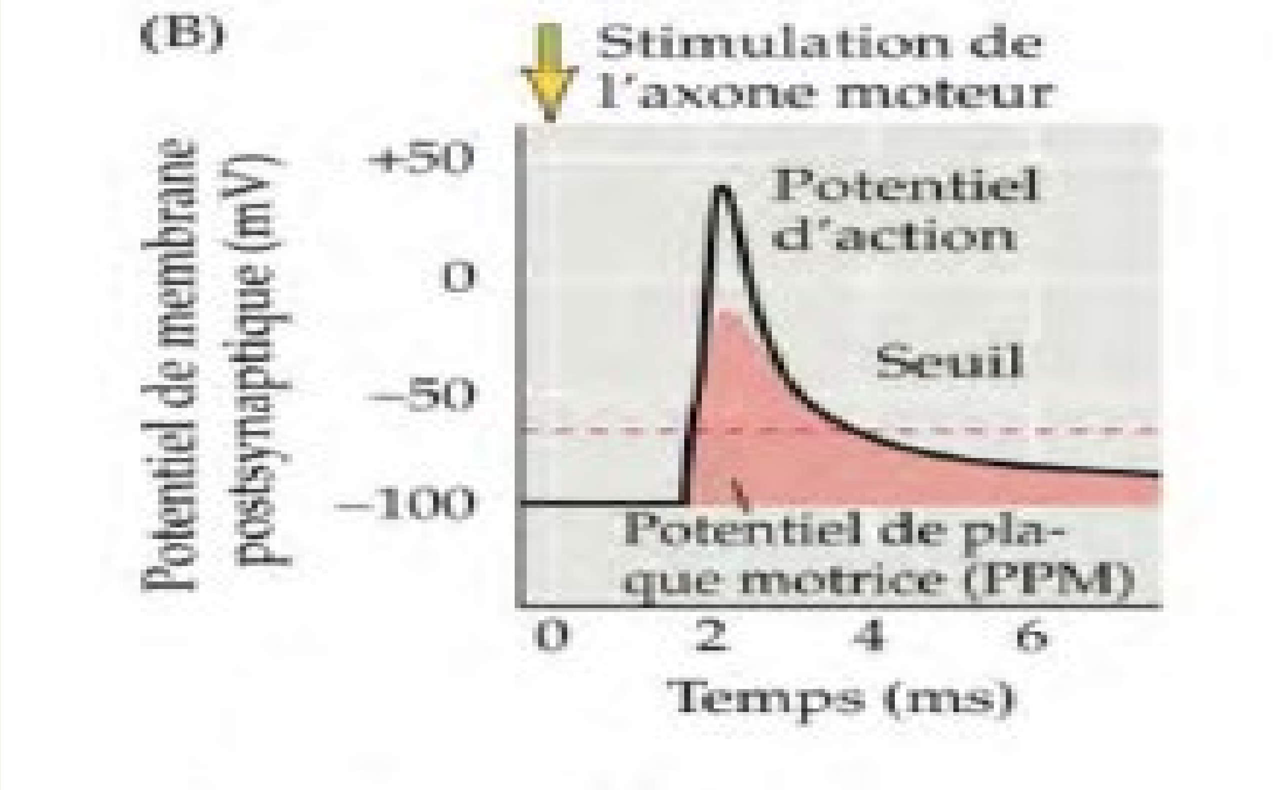
Motor End-Plate Action Potential

In the normal state, stimulation of the motor nerve always induces an Action Potential (AP) that propagates across the muscle membrane.

After curarization or blockade by TTX, a local depolarization can be isolated: the End-Plate Potential (EPP).

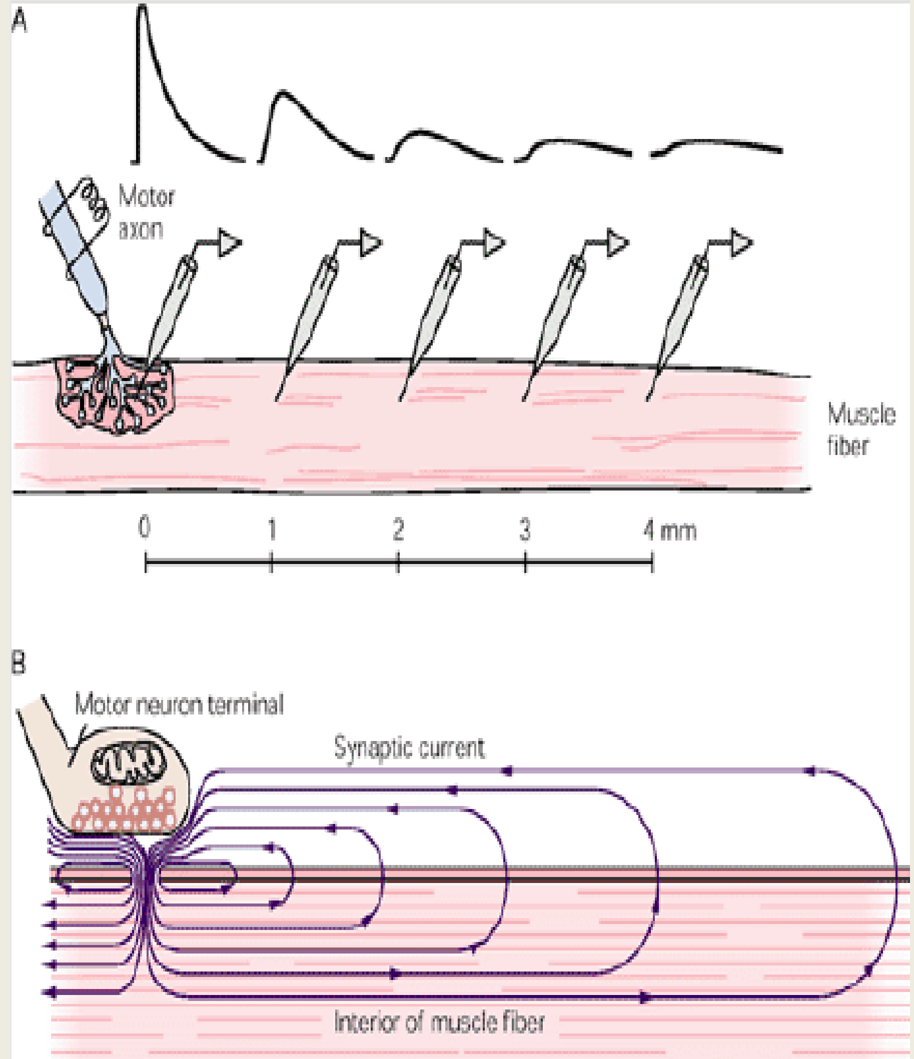


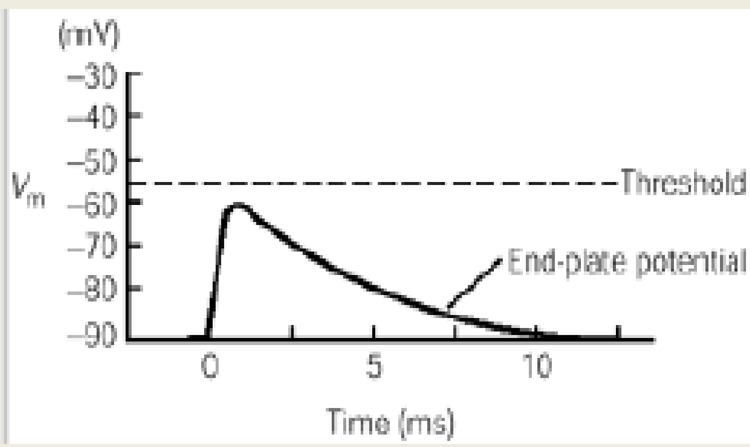

(A)
Stimulation
Stimulaion
de l'axone
Enregistrement
Enregistrement
du potentiel
de membrane
postsynaptique
Axone
Cellule musculaire


(B)
Stimulation de
l'axone moteur
Potentiel de membrane
postsynaptique (mV)
+50
0
−50
−100
Potentiel
d'action
Seuil
Potentiel de pla-
que motrice (PPM)
0
2
4
6
Temps (ms)

Motor End-Plate Action Potential

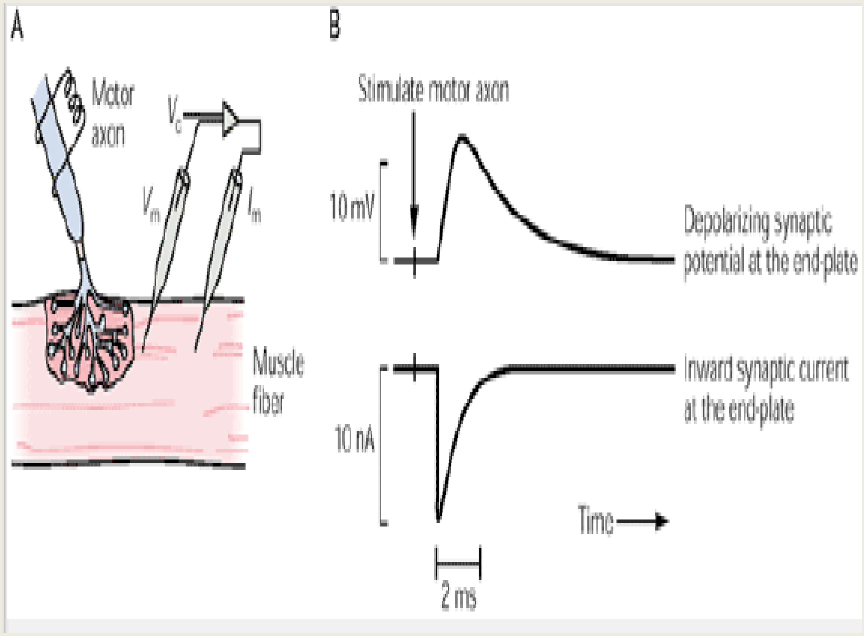
It is a local phenomenon: reaches its maximum in 02 msec and then decreases exponentially to cancel in 20 msec.





Ionic mechanisms of the Motor End-Plate Action Potential (MEPAP)

VOLTAGE CLAMP EXPERIMENT
A decrease in membrane resistances is observed with an increase in permeability and ion passage.



results from a current
Ample and brief entry
Depolarizing (Na+) followed by a

Low and slow outgoing current
Repolarizing (K+).

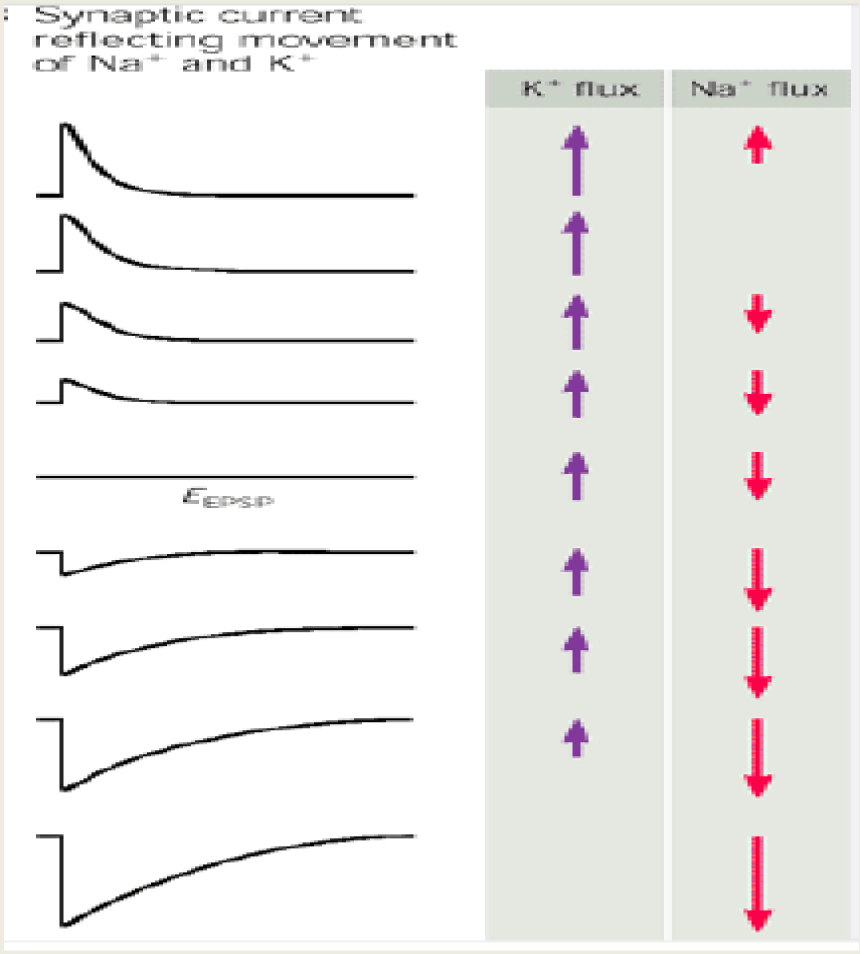
75
mV

66
40
+20
0

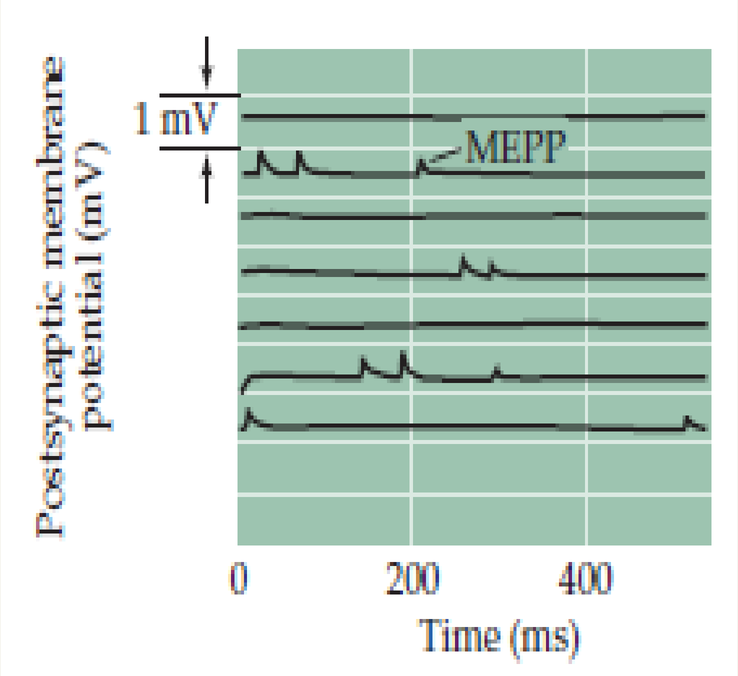
-20
-50

-90

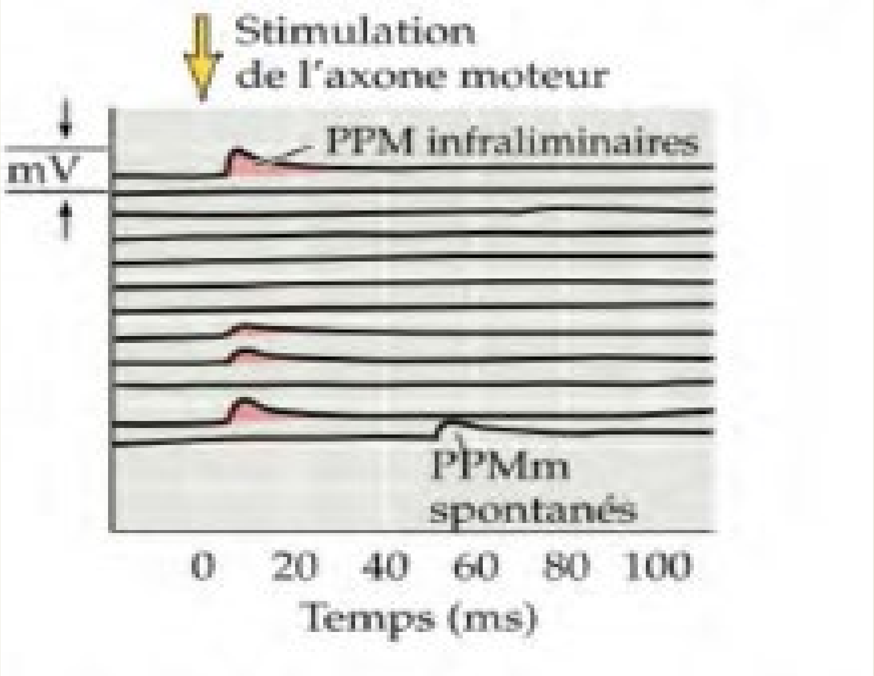
-100



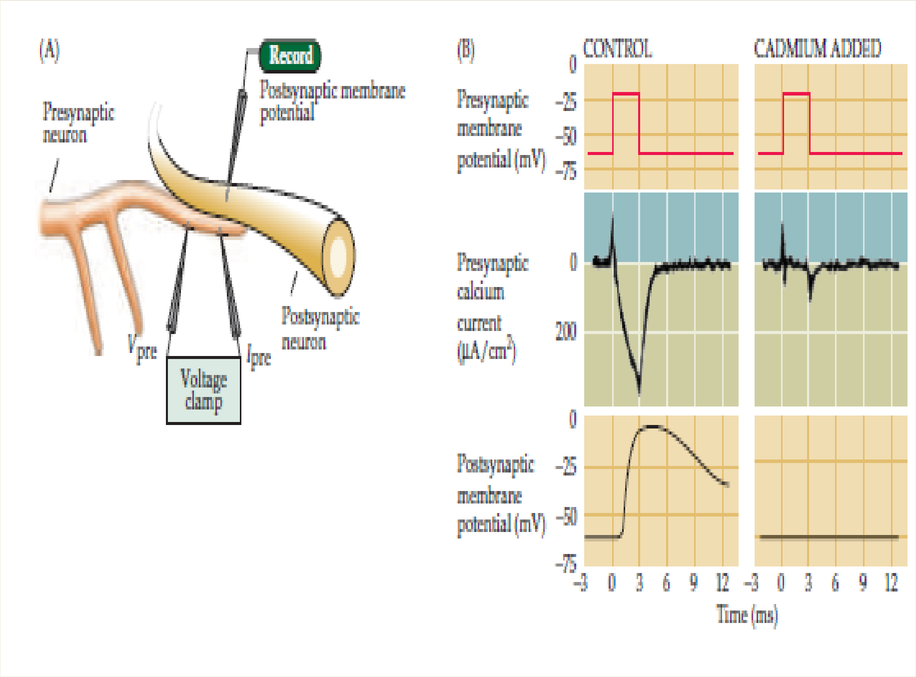
Motor End-Plate Action Potential (MEPAP) Miniatures



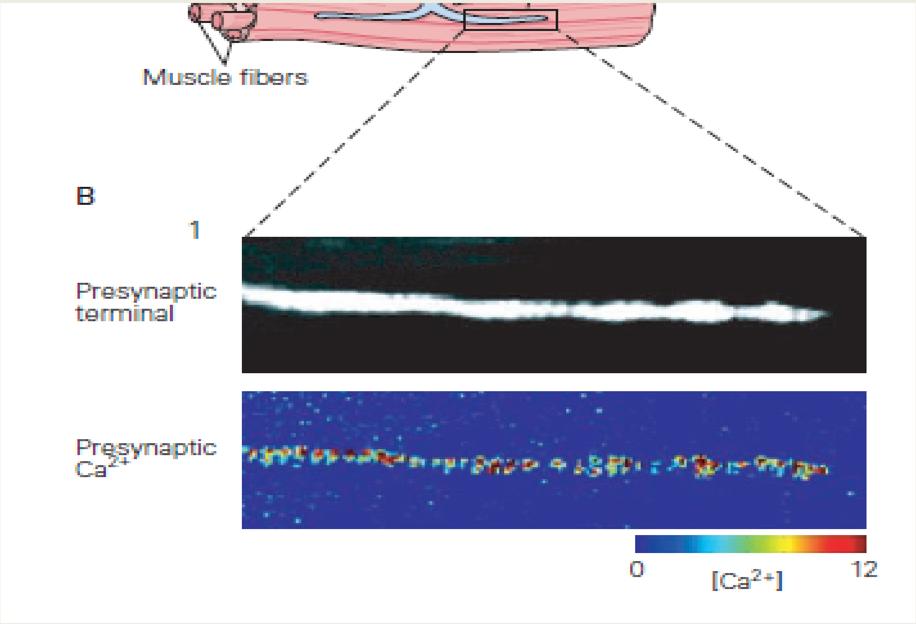
Registered outside of any
 stimulation and would be the
result
of the liberation of Quanta
acetylcholine



Role of calcium in electrosecretion



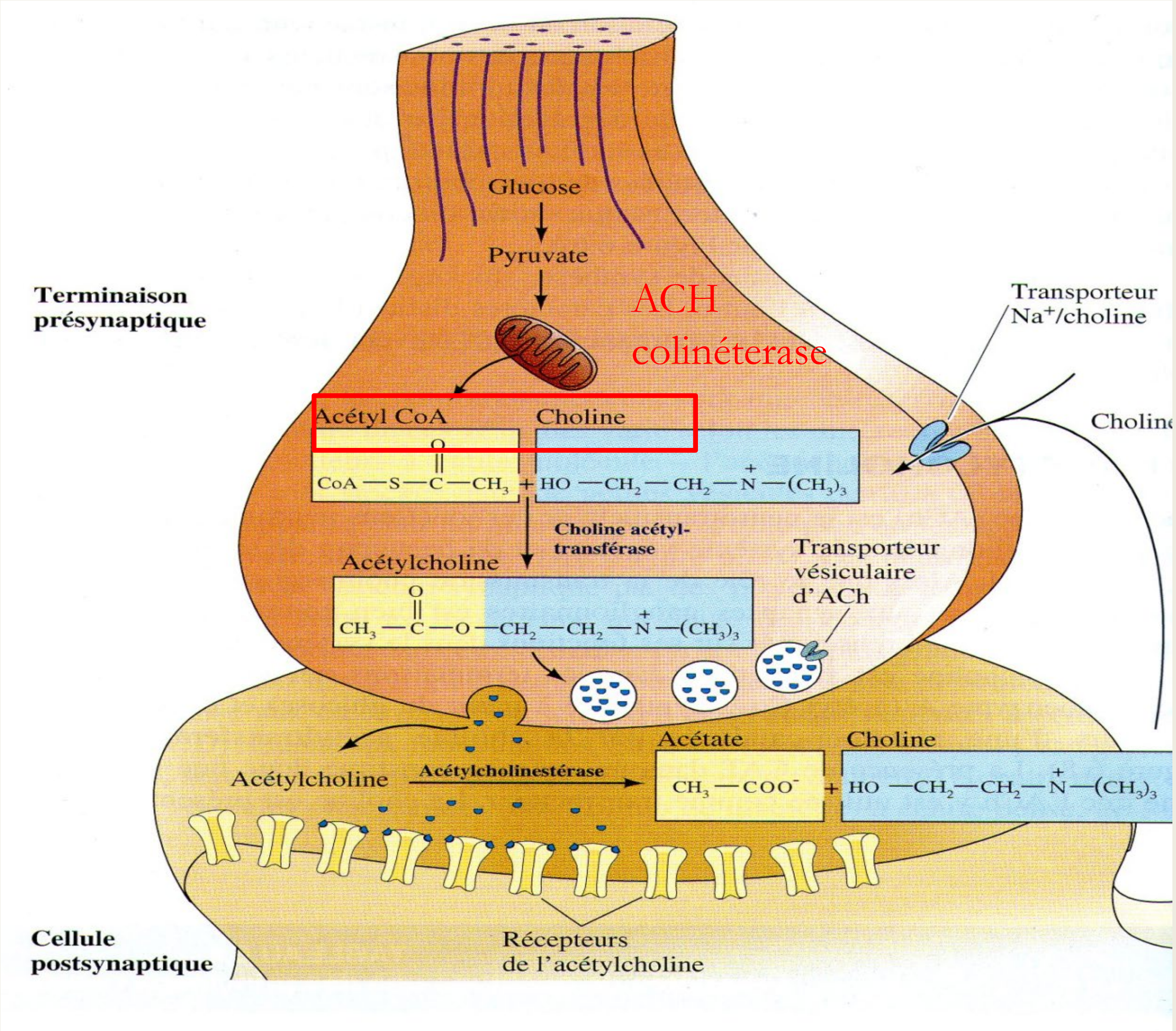
Fluorescence measurement (fura-2) during repetitive presynaptic stimulation of a neuromuscular synapse.



BIOCHEMICAL PHENOMENA

Neurotrametter: Acetylcholine (ACh): This neurotransmitter is an amine

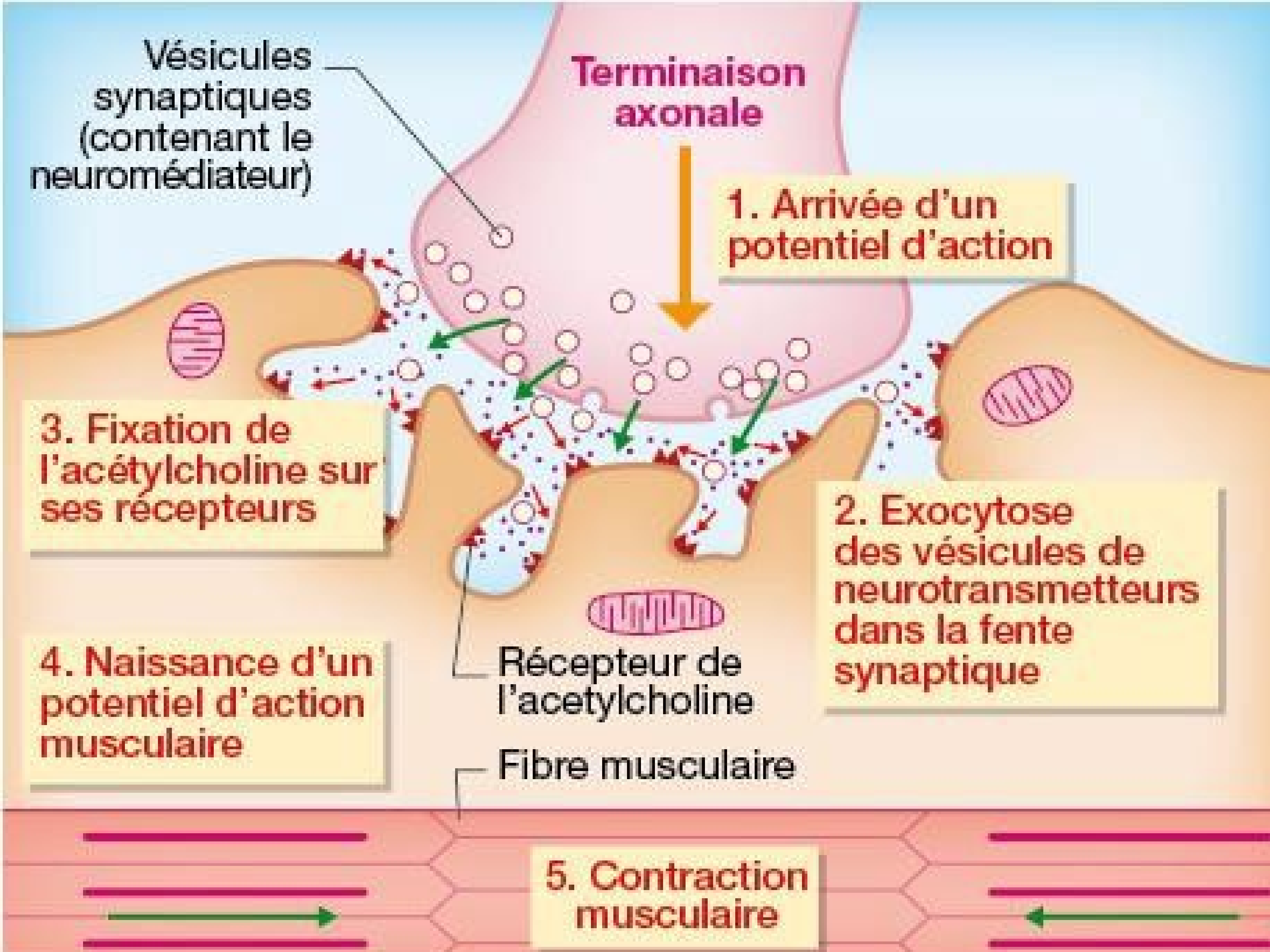
- cycle of l'acetylcholine



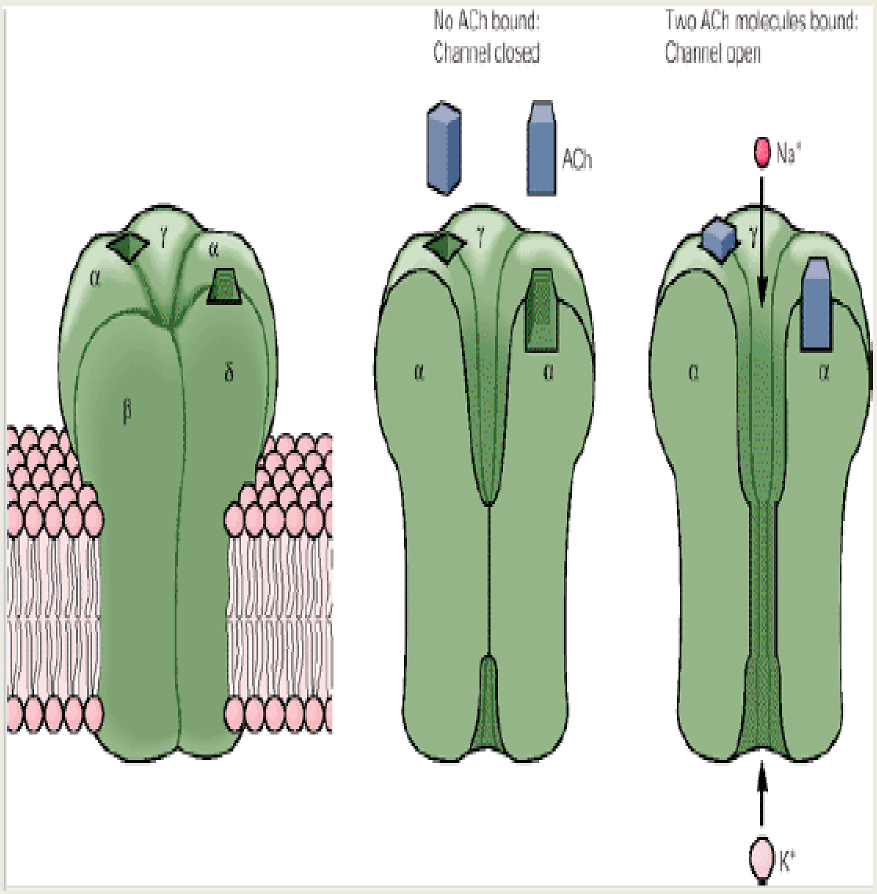
- **Synthesis:** ACh is synthesized in the nerve terminals of cholinergic fibers, largely within the **axoplasm**, from **acetyl-coenzyme A (acetyl-CoA)** and **choline**. This reaction is catalyzed by **choline acetyltransferase (ChAT)**.
- **Storage:** ACh is transferred and stored inside **vesicles**.
- **Release.**
- **Binding to receptors.**
- **Inactivation:** Unlike other small-molecule neurotransmitters, its postsynaptic action is not terminated by reuptake but by **hydrolysis**, which is carried out by **acetylcholinesterase (AChE)**.
- The released **choline** is then taken back up (**reuptaken**) into the presynaptic terminal.

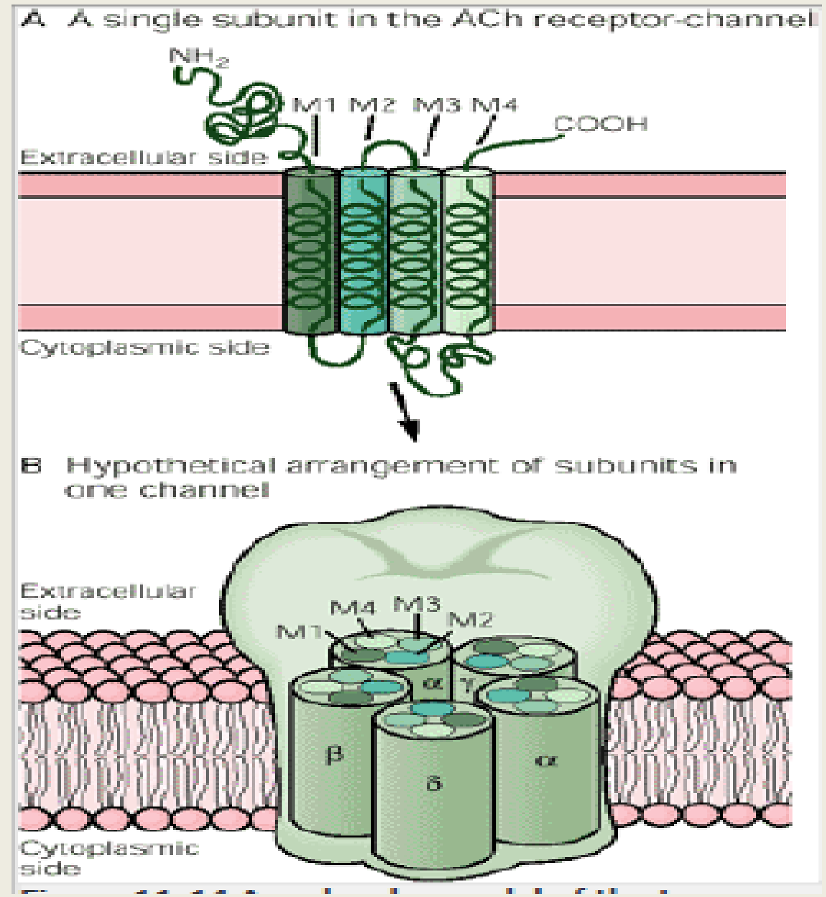
Functioning of the the Neuromuscular Junction –NMJ-

- When the nerve is stimulated, an Action Potential (AP) travels along the nerve to its nerve terminals.
- Presynaptic depolarization causes the brief and rapid opening of voltage-gated Ca^{2+} channels.
- Influx of extracellular Ca^{2+} into the nerve terminals. Release of ACh in multimolecular "packets" (quanta).
- Binding of ACh to receptors located on the postsynaptic membrane. Opening of Na^+ channels at the motor end-plate, producing a brief but significant ionic current and the depolarization of the postsynaptic membrane (Motor End-Plate Potential).
- If its amplitude is sufficient, it brings the muscle cell membrane potential above the threshold for firing a postsynaptic
- Action Potential, which in turn triggers the contraction of the muscle fiber.


Vésicules synaptiques (contenant le neuromédiateur)
Terminaison axonale
1. Arrivée d'un potentiel d'action
3. Fixation de l'acétylcholine sur ses récepteurs
2. Exocytose des vésicules de neurotransmetteurs dans la fente synaptique
4. Naissance d'un potentiel d'action musculaire
Récepteur de l'acetylcholine
Fibre musculaire
5. Contraction musculaire

Récepteur de la plaque motrice





PHARMACOLOGY OF THE NEUROMUSCULAR JUNCTION:

Neuromuscular transmission can be blocked via three pathways:Inhibition of ACh synthesis.Inhibition of ACh release .Interference with the postsynaptic action of ACh.

1 - Agents inhibiting ACh synthesis: Hemicholinium: a competitive inhibitor of choline. Vesamicol: blocks the transport of ACh into the vesicles.

2 - Agents blocking ACh release: Inhibitors of calcium entry: Mg^{2+}, aminoglycoside antibiotics (streptomycin, neomycin), and calcium antagonists. Neurotoxins: botulinum toxin, beta-bungarotoxin.

3 - Postsynaptic agents:

3-1 - Non-depolarizing blocking agents:

They are competitive antagonists of ACh at the motor end-plate, acting by decreasing the number of sites accessible to ACh. We distinguish between:Reversible competitive antagonists: Curare.

Irreversible competitive antagonists: \alpha-bungarotoxin.

3-2 - Depolarizing blocking agents:

These are ACh agonists, for example: decamethonium, suxamethonium.

4 - Anticholinesterases: Used in the treatment of myasthenia gravis (neostigmine, pyridostigmine).

Pathologies of the Neuromuscular Junction

Myasthenic syndromes:

presynaptic myasthenic syndromes

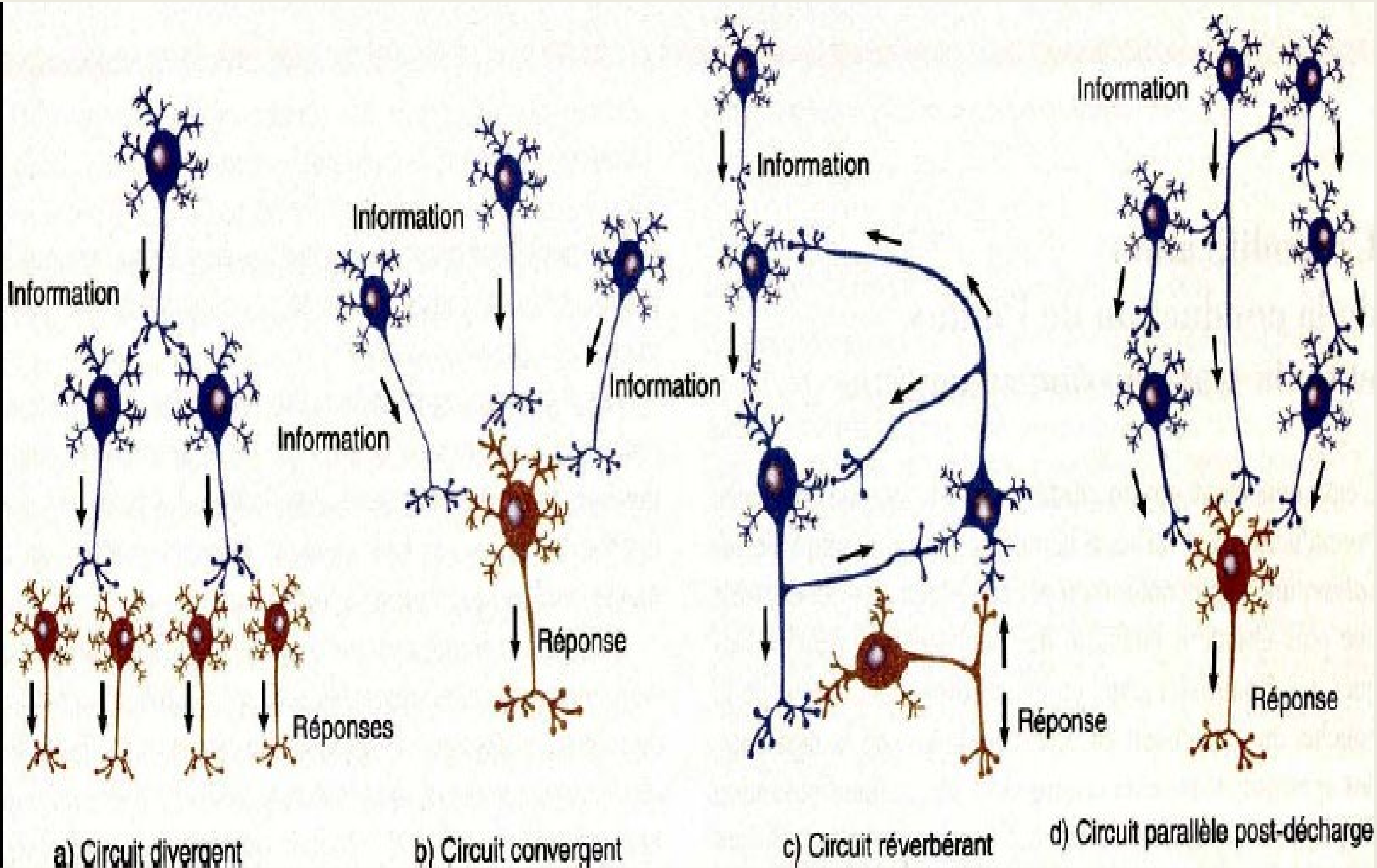
Postsynaptic myasthenic syndrome

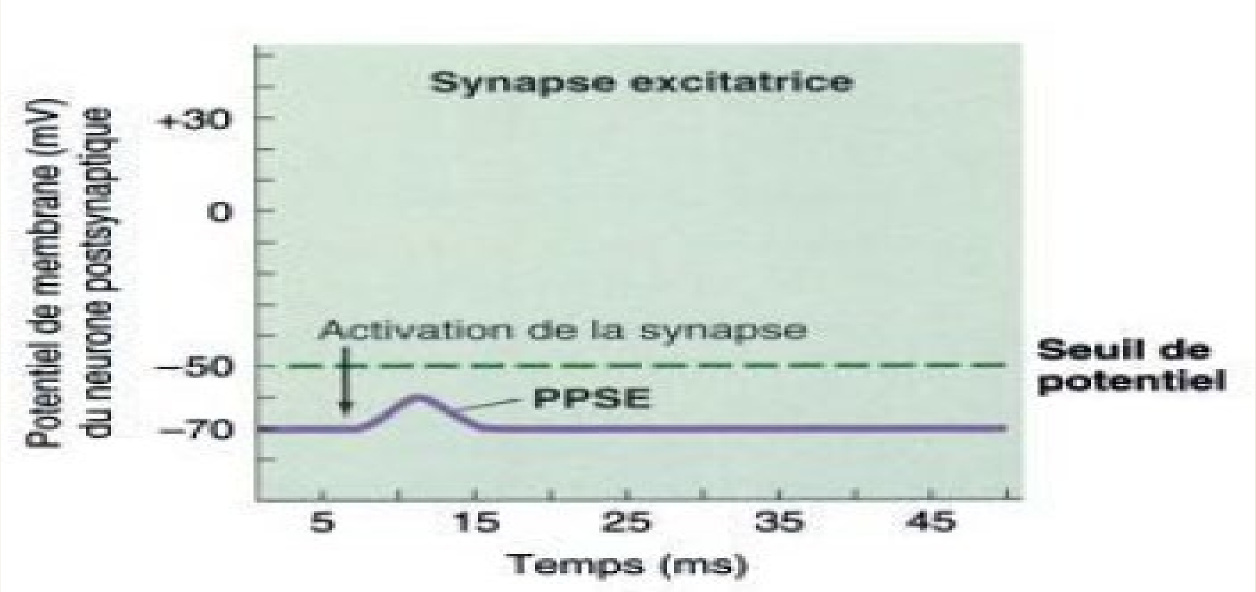
- Le Syndrome de LAMBERT EATON
- Botulism
- Congenital myasthenic syndromes
- Venomous Poisoning
- Chemical Poisoning

Chemical poisoning

- A large number of chemical substances capable of blocking the **Neuromuscular Junction (NMJ)** have been synthesized, initially within the framework of research on chemical warfare.
- Numerous **insecticides** and **nerve gases**, including **mustard gas** (used during World War I) and **Sarin** (a neurotoxic gas).
- **Organophosphates**.
- These are **irreversible inhibitors of AChE** (Acetylcholinesterase).
- They prevent the degradation of **ACh** (Acetylcholine) and lead to an accumulation of ACh within the synapses.
- **Desensitization of receptors**: despite the presence of ACh, the neurotransmitter-sensitive channels close.
- The excess of ACh depolarizes the postsynaptic membrane and renders it **refractory** to any further release of ACh. This state of desensitization can persist for several seconds after the removal of the neurotransmitter and causes **neuromuscular paralysis**.
- **Curare** is an inhibitor of synaptic transmission that acts by binding to receptors, thereby blocking the normal action of the neurotransmitter.

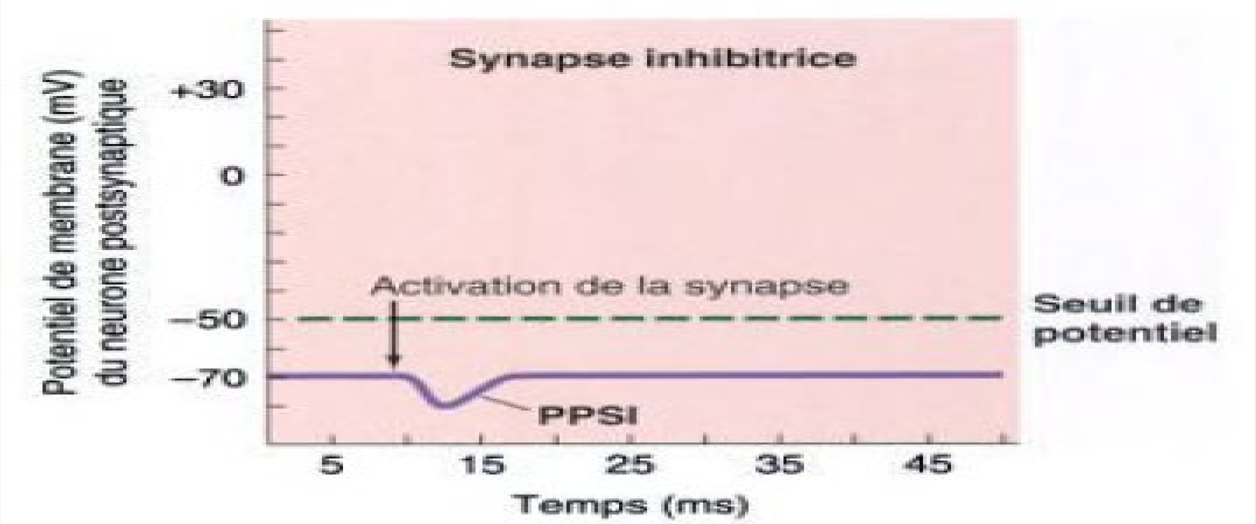
Neuro-neuronal synapse





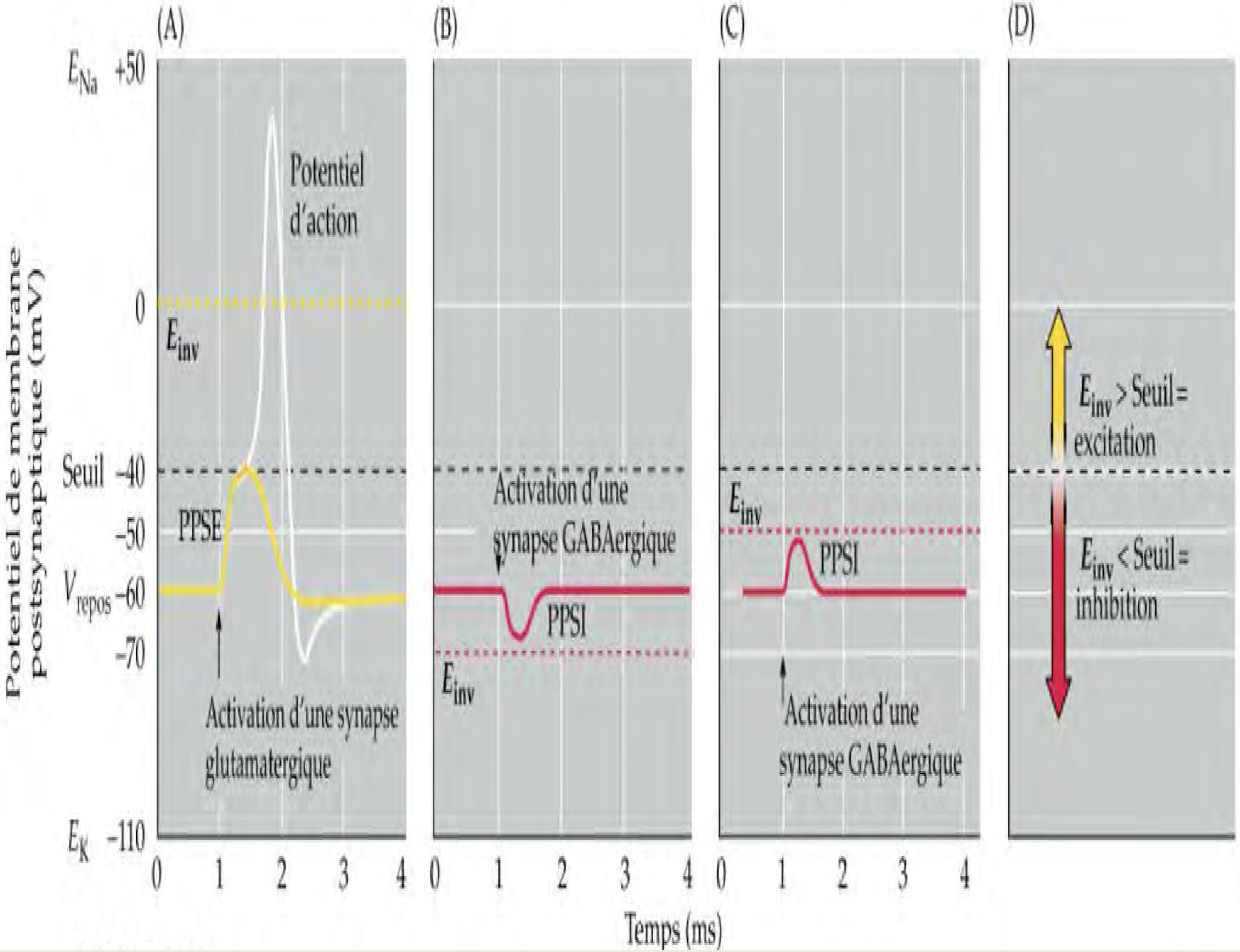
Na++ entry

Potentiel post synaptique excitateur



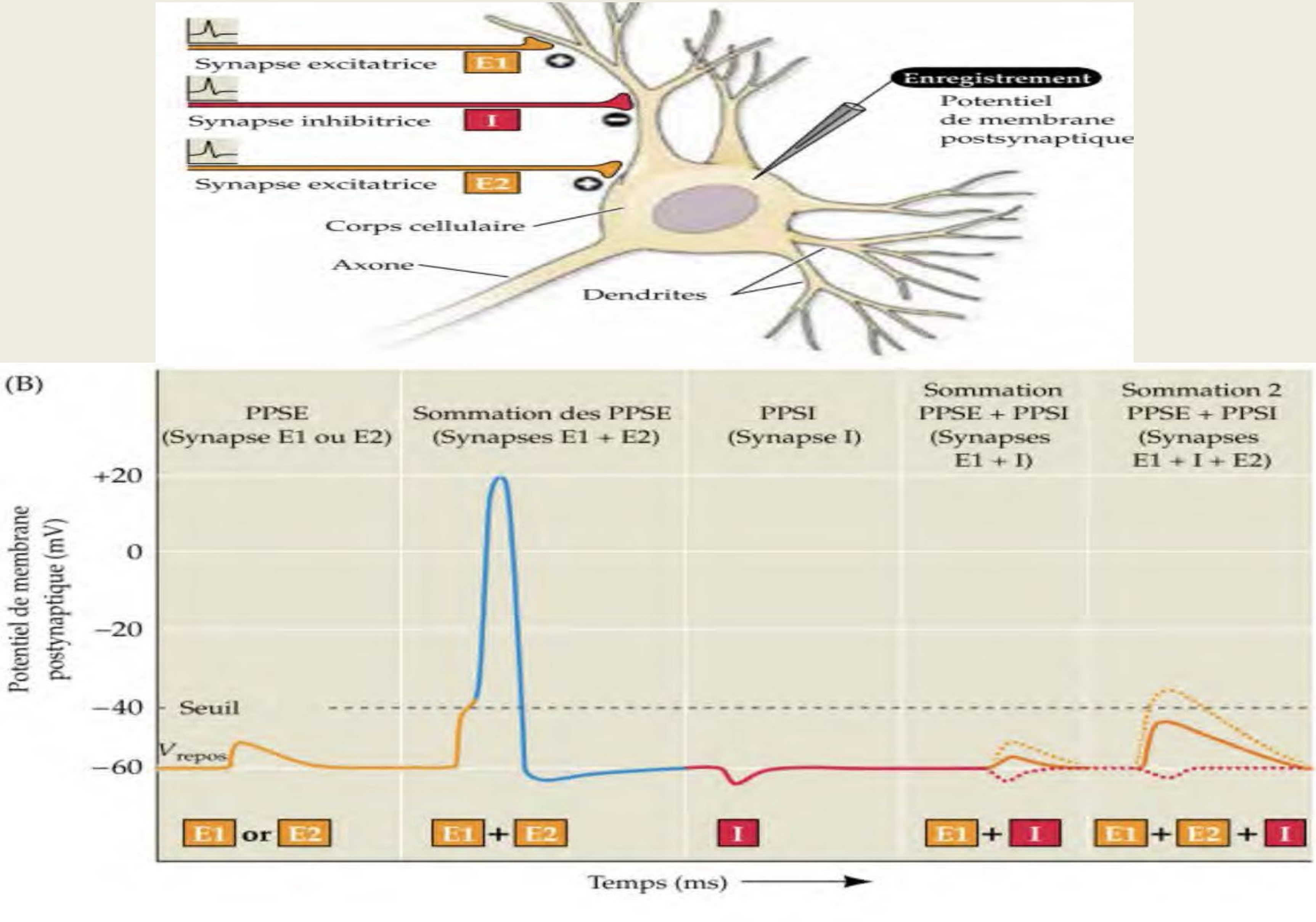
CL- entry

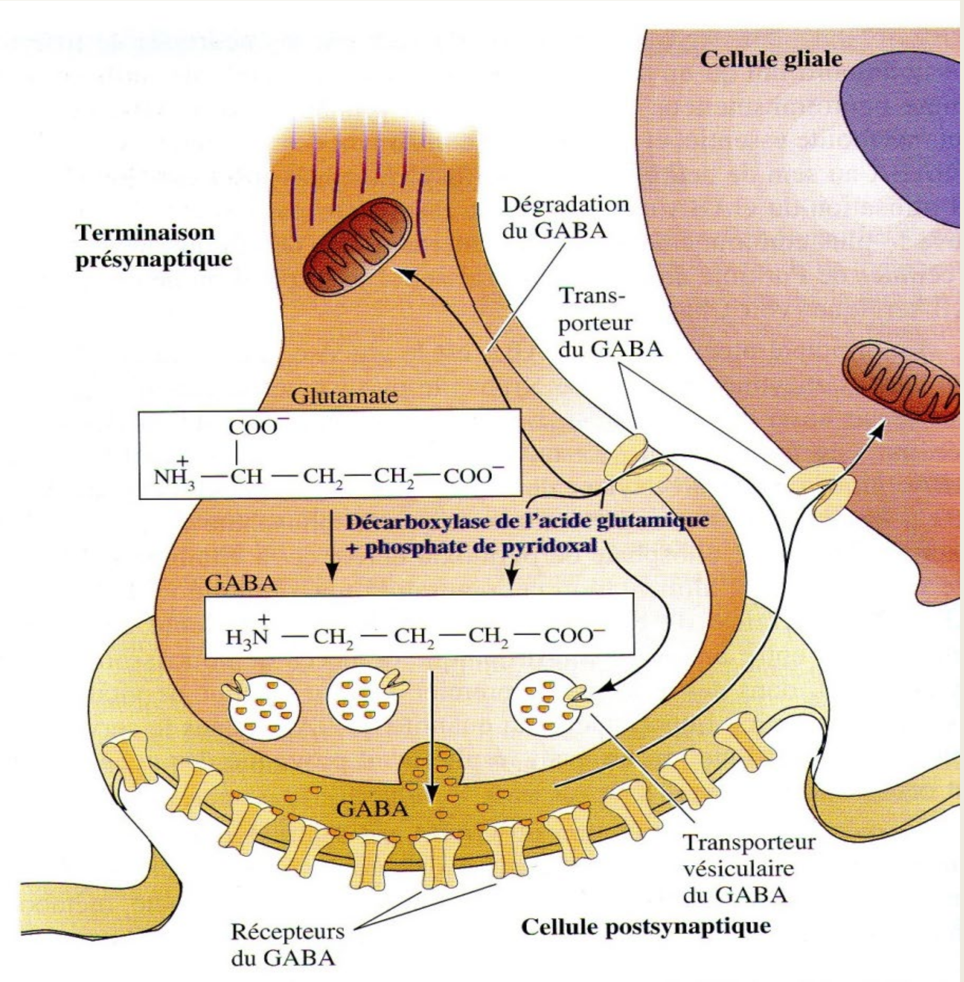
Potentiel post synaptique inhibiteur

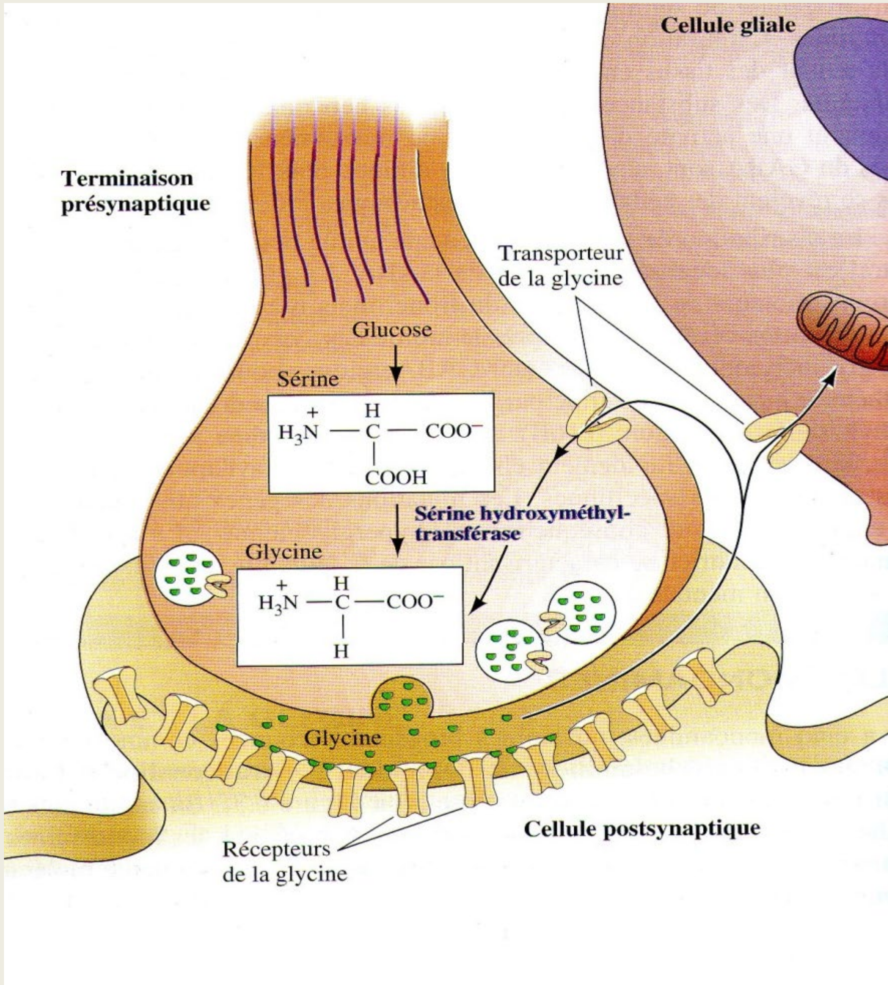


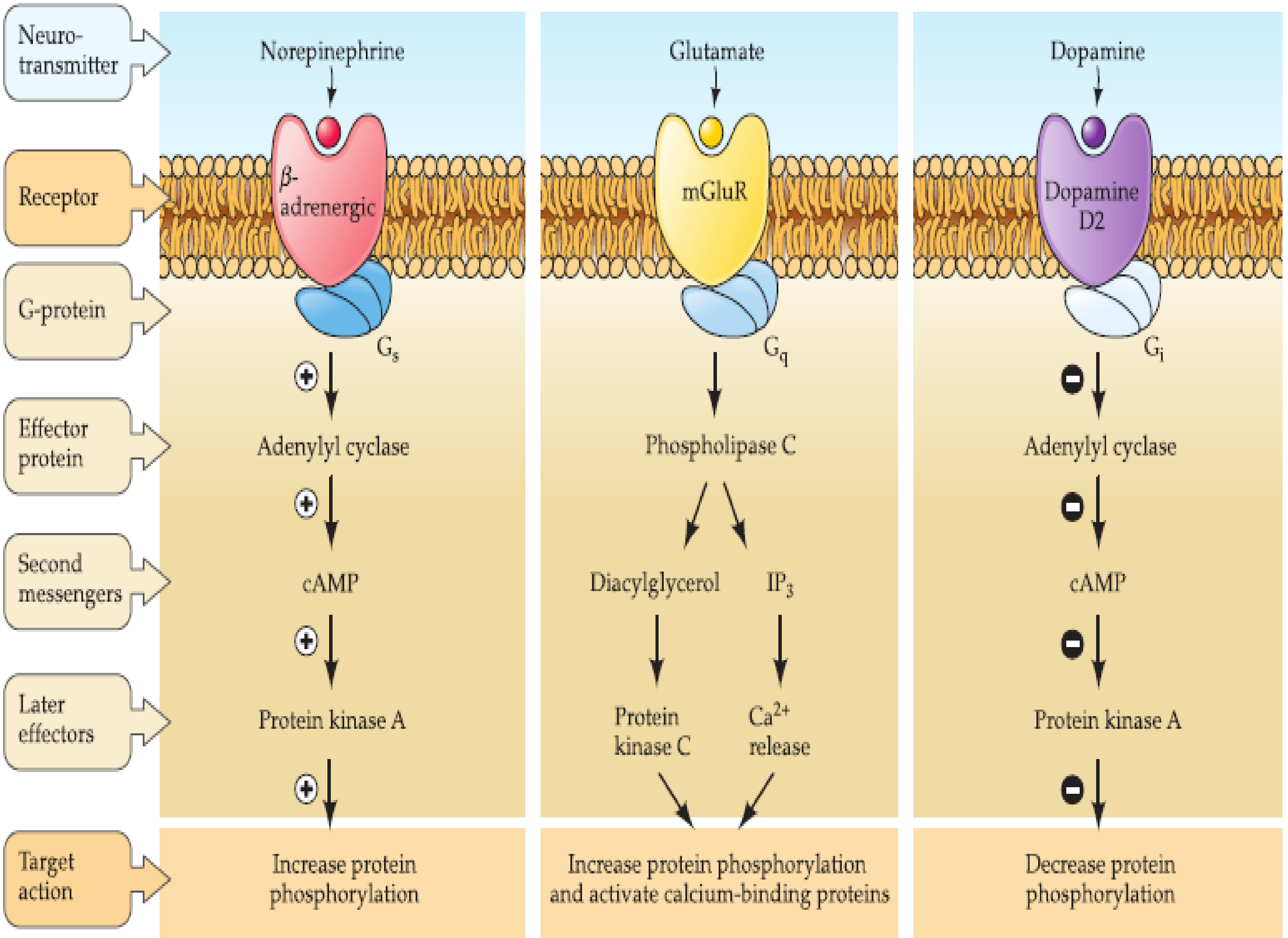
(A)
(B)
(C)
(D)
E_{Na} +50
0
Seuil −40
−50
V_{repos} −60
−70
E_K −110
Potentiel de membrane postsynaptique (mV)
Potentiel d'action
E_{inv}
PPSE
Activation d'une synapse glutamatergique
Activation d'une synapse GABAergique
PPSI
E_{inv}
E_{inv}
PPSI
Activation d'une synapse GABAergique
E_{inv} > Seuil = excitation
E_{inv} < Seuil = inhibition
0 1 2 3 4
Temps (ms)

Spatial and temporal summation




Cellule gliale
Terminaison présynaptique
Dégradation du GABA
Trans-porteur du GABA
Glutamate
COO⁻
NH3 — CH — CH2 — CH2 — COO⁻
Décarboxylase de l'acide glutamique + phosphate de pyridoxal
GABA
H3N — CH2 — CH2 — CH2 — COO⁻
GABA
Transporteur vésiculaire du GABA
Récepteurs du GABA
Cellule postsynaptique


Cellule gliale
Terminaison présynaptique
Transporteur de la glycine
Glucose
Sérine
H3N — C — COO⁻ H COOH
Sérine hydroxyméthyl-transférase
Glycine
H3N — C — COO⁻ H H
Glycine
Cellule postsynaptique
Récepteurs de la glycine



Neuro-
transmitter
Receptor
G-protein
Effector
protein
Second
messengers
Later
effectors
Target
action
Norepinephrine
β-
adrenergic
G_s
Adenylyl cyclase
cAMP
Protein kinase A
Increase protein
phosphorylation
Glutamate
mGluR
G_q
Phospholipase C
Diacylglycerol
IP_3
Protein
kinase C
Ca^{2+}
release
Increase protein phosphorylation
and activate calcium-binding proteins
Dopamine
Dopamine
D2
G_i
Adenylyl cyclase
cAMP
Protein kinase A
Decrease protein
phosphorylation

Neurotransmitters

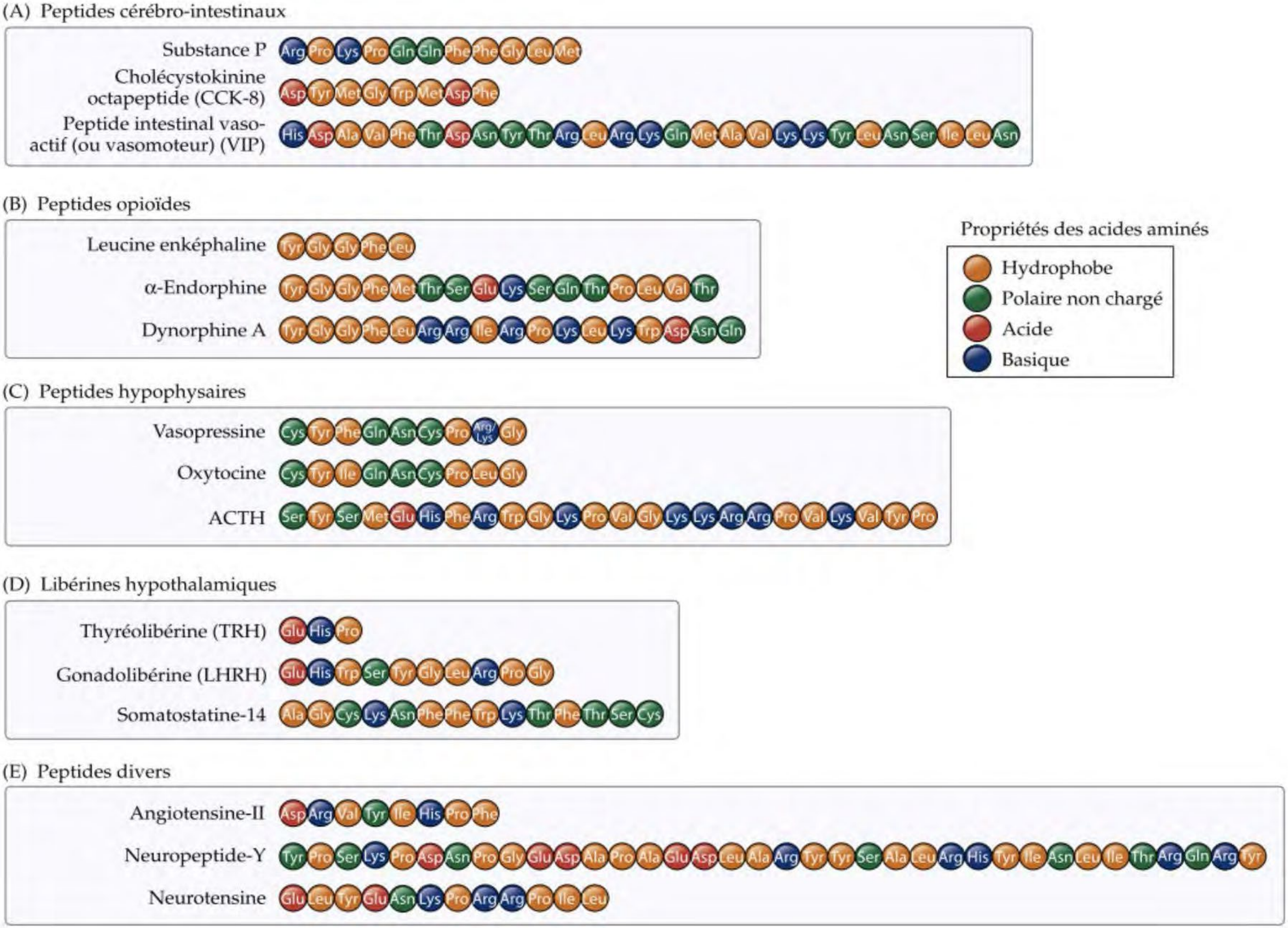
Les neurotransmetteurs

Table 13–1 Small-Molecule Transmitter Substances and Their Precursors

Transmitter	Precursor
Acetylcholine	Choline
Biogenic amines	
Dopamine	Tyrosine
Norepinephrine	Tyrosine
Epinephrine	Tyrosine
Serotonin	Tryptophan
Histamine	Histidine
Melatonin	Serotonin
Amino acids	
Aspartate	Oxaloacetate
γ-Aminobutyric acid	Glutamine
Glutamate	Glutamine
Glycine	Serine
ATP	ADP
Adenosine	ATP
Arachidonic acid	Phospholipids
Carbon monoxide	Heme
Nitric oxide	Arginine

Les neurotransmetteurs.

Classe I
Acétylcholine
Classe II : les amines
Noradrénaline Adrénaline Dopamine Sérotonine Histamine
Classe III : les acides aminés
Acide γ-aminobutyrique (GABA) Glycine Glutamate Aspartate et NMDA
Classe IV : les peptides (actuellement, plus de 100 peptides sont considérés comme neurotransmetteurs)
A. Facteurs hypothalamiques de libération hormonale Stimuline de l'hormone thyréotrope Stimuline de l'hormone lutéinisante Somatostatine (facteur inhibiteur de l'hormone de croissance)
B. Peptides hypophysaires ACTH (hormone corticotrope) β-endorphine Stimuline des μ-mélanocytes Vasopressine Ocytocine
C. Peptides agissant sur le système digestif et le cerveau Leucine enképhaline Méthionine enképhaline Substance P Cholécystokinine Polypeptide intestinal vasoactif (VIP) Neurotensine Insuline Glucagon
D. D'autres tissus Angiotensine II Bradykinine Carnosine Bombésine


(A) Peptides cérébro-intestinaux
Substance P
Arg Pro Lys Pro Gln Gln Phe Phe Gly Leu Met
Cholécystokinine octapeptide (CCK-8)
Asp Tyr Met Gly Trp Met Asp Phe
Peptide intestinal vaso-actif (ou vasomoteur) (VIP)
His Asp Ala Val Phe Thr Asp Asn Tyr Thr Arg Leu Arg Lys Gln Met Ala Val Lys Lys Tyr Leu Asn Ser Ile Leu Asn
(B) Peptides opioïdes
Leucine enképhaline
Tyr Gly Gly Phe Leu
α-Endorphine
Tyr Gly Gly Phe Met Thr Ser Glu Lys Ser Gln Thr Pro Leu Val Thr
Dynorphine A
Tyr Gly Gly Phe Leu Arg Arg Ile Arg Pro Lys Leu Lys Trp Asp Asn Gln
(C) Peptides hypophysaires
Vasopressine
Cys Tyr Phe Gln Asn Cys Pro Arg/Lys Gly
Oxytocine
Cys Tyr Ile Gln Asn Cys Pro Leu Gly
ACTH
Ser Tyr Ser Met Glu His Phe Arg Trp Gly Lys Pro Val Gly Lys Lys Arg Arg Pro Val Lys Val Tyr Pro
(D) Libérines hypothalamiques
Thyréolibérine (TRH)
Glu His Pro
Gonadolibérine (LHRH)
Glu His Trp Ser Tyr Gly Leu Arg Pro Gly
Somatostatine-14
Ala Gly Cys Lys Asn Phe Phe Trp Lys Thr Phe Thr Ser Cys
(E) Peptides divers
Angiotensine-II
Asp Arg Val Tyr Ile His Pro Phe
Neuropeptide-Y
Tyr Pro Ser Lys Pro Asp Asn Pro Gly Glu Asp Ala Pro Ala Glu Asp Leu Ala Arg Tyr Tyr Ser Ala Leu Arg His Tyr Ile Asn Leu Ile Thr Arg Gln Arg Tyr
Neurotensine
Glu Leu Tyr Glu Asn Lys Pro Arg Arg Pro Ile Leu
Propriétés des acides aminés
Hydrophobe
Polaire non chargé
Acide
Basique

Neuro-neuronal synapses exhibit specific characteristics:

They ensure extremely rapid communication.They provide specific and modulable communication between neurons.

Their plasticity, through phenomena such as facilitation, synaptic depression, or Long-Term Potentiation (LTP), constitutes the physiological basis for:

Learning processes.Memory. Adaptations of the nervous system to environmental stimuli.

Neuro-glandular synapses: Example synapses of the autonomic nervous system.

Refer to the course on the autonomic or visceral nervous system

- The synapse is not merely a simple point of transmission, but a dynamic and regulatory structure, essential for information processing and the integrated functioning of the brain

- Beyond the central and somatic nervous systems, synapses play a role in information transmission within the autonomic nervous system, which is involved in major regulatory processes and the regulation of visceral functions

- Similarly, at the level of the motor end-plate, the synapse represents a highly specialized interface between the motor neuron and the muscle.